



NTNU – Trondheim
Norwegian University of
Science and Technology

TET 4565 -Electricity Markets and Energy System Planning, Specialisation Course

**Decomposition techniques, Bender's
decomposition (theory and application)**

Special thanks to professor Jalal Kazempour at DTU

Hossein Farahmand

Professor

Autumn 2026

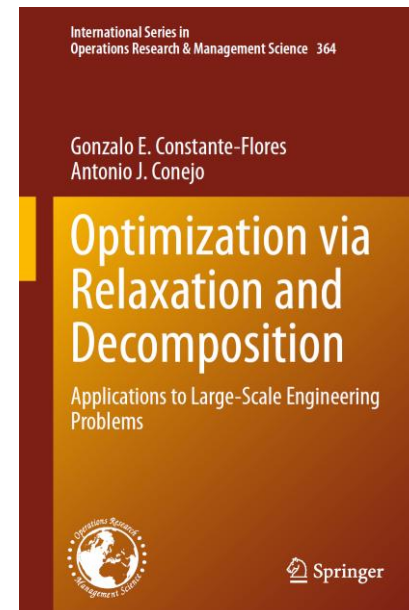
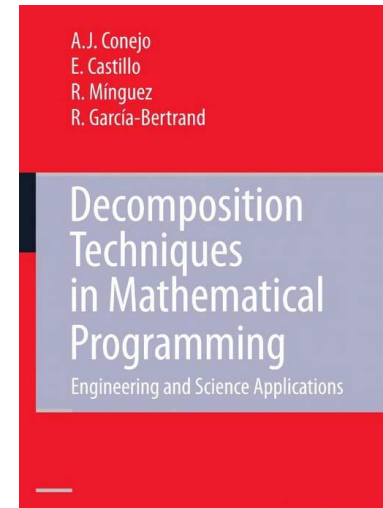
Overview

- Decomposition techniques
- Explain the functioning of Benders' decomposition (concept)
- Implement it to an illustrative problem
- **Implementation of Bender's decomposition to two-stage stochastic programming**

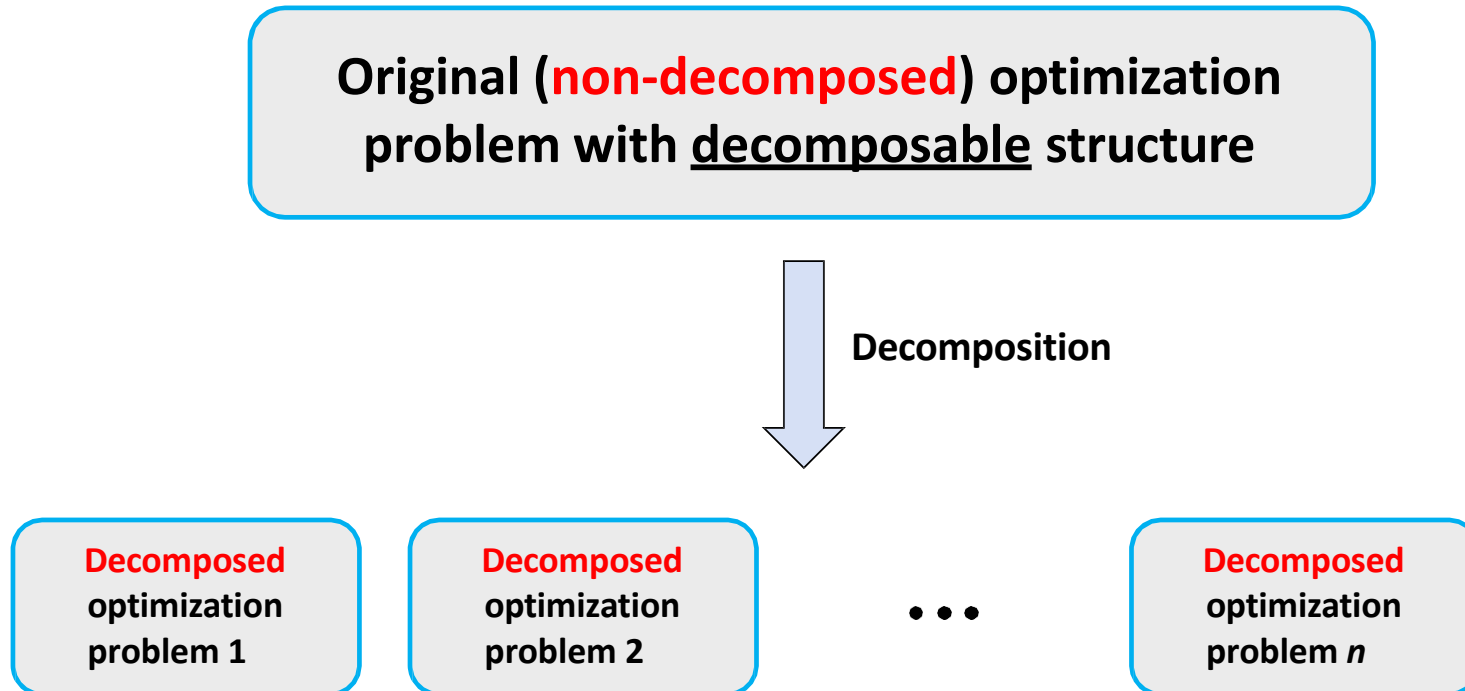
Reference

A. J. Conejo, E. Castillo, R. Minguez, and R. Garcia-Bertrand, *Decomposition Techniques in Mathematical Programming: Engineering and Science Applications*. Berlin, Germany: Springer, 2006.
(Chapter 3, 3.3)

Chapter 5
Solving Optimization Problems with
Complicating Variables



Decomposition: main idea



Each decomposed problem is **easier to solve** than the original (non-decomposed) problem!

Decomposable structure

Optimization problems with decomposable structure:

- Problems with complicating variable(s):

The original problem is decomposed if **complicating variables** are **fixed** to given values!

- Problems with complicating constraint(s):

The original problem is decomposed if **complicating constraints** are **relaxed** (removed)!

In the literature, “complicating” variables/constraints are also called as “coupling” or “linking” variables/constraint!

Features of decomposition techniques

- Iterative solution techniques
- Original optimization problem decomposes to:
 - A single **master** problem (not necessarily an optimization problem)
 - A set of **subproblems**

Illustrative example

Single-node single-year (static) generation expansion problem, considering two existing generators (units 1 and 2) and one candidate generator (unit 3)

Linking variable or complicating variable

$$\text{Minimise } \underbrace{c_3^{inv} x_3}_{\text{Expansion cost of candidate unit}} + \underbrace{\sum_h [c_1 g_{h,1} + c_2 g_{h,2} + c_3 g_{h,3}]}_{\text{Operational cost of (existing and candidate) units}}$$

subject to

$$0 \leq g_{h,1} \leq \bar{g}_1 \quad \forall h$$

$$0 \leq g_{h,2} \leq \bar{g}_2 \quad \forall h$$

$$0 \leq g_{h,3} \leq x_3 \quad \forall h$$

$$g_{h,1} + g_{h,2} + g_{h,3} = D_{h,1} \quad \forall h$$

$$x_3 \geq 0$$

Illustrative example- decomposed problem

We fixed $x_3 \rightarrow x_3^{fixed}$ and remove the fixed term from the objective function.
The subproblems will be:

$$\left\{ \begin{array}{l} \text{Minimise } c_1 g_{h,1} + c_2 g_{h,2} + c_3 g_{h,3} \\ \quad g_{h,1}, g_{h,2}, g_{h,3} \\ \text{subject to,} \\ 0 \leq g_{h,1} \leq \bar{g}_1 \\ 0 \leq g_{h,2} \leq \bar{g}_2 \\ 0 \leq g_{h,3} \leq x_3^{fixed} \\ g_{h,1} + g_{h,2} + g_{h,3} = D_{h,1} \end{array} \right\} \forall h$$

Seminal paper of Benders

Numerische Mathematik 4, 238—252 (1962)

Partitioning procedures for solving mixed-variables programming problems*

By

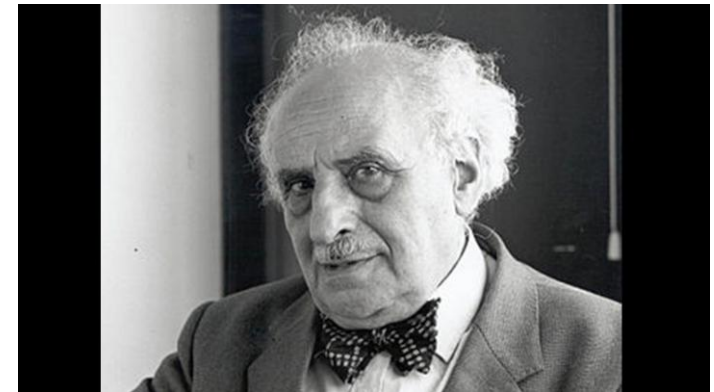
J. F. BENDERS**

I. Introduction

In this paper two slightly different procedures are presented for solving mixed-variables programming problems of the type

$$\max \{c^T x + f(y) \mid Ax + F(y) \leq b, x \in R_p, y \in S\}, \quad (1.1)$$

where $x \in R_p$ (the p -dimensional Euclidean space), $y \in R_q$, and S is an arbitrary subset of R_q . Furthermore, A is an (m, p) matrix, $f(y)$ is a scalar function and



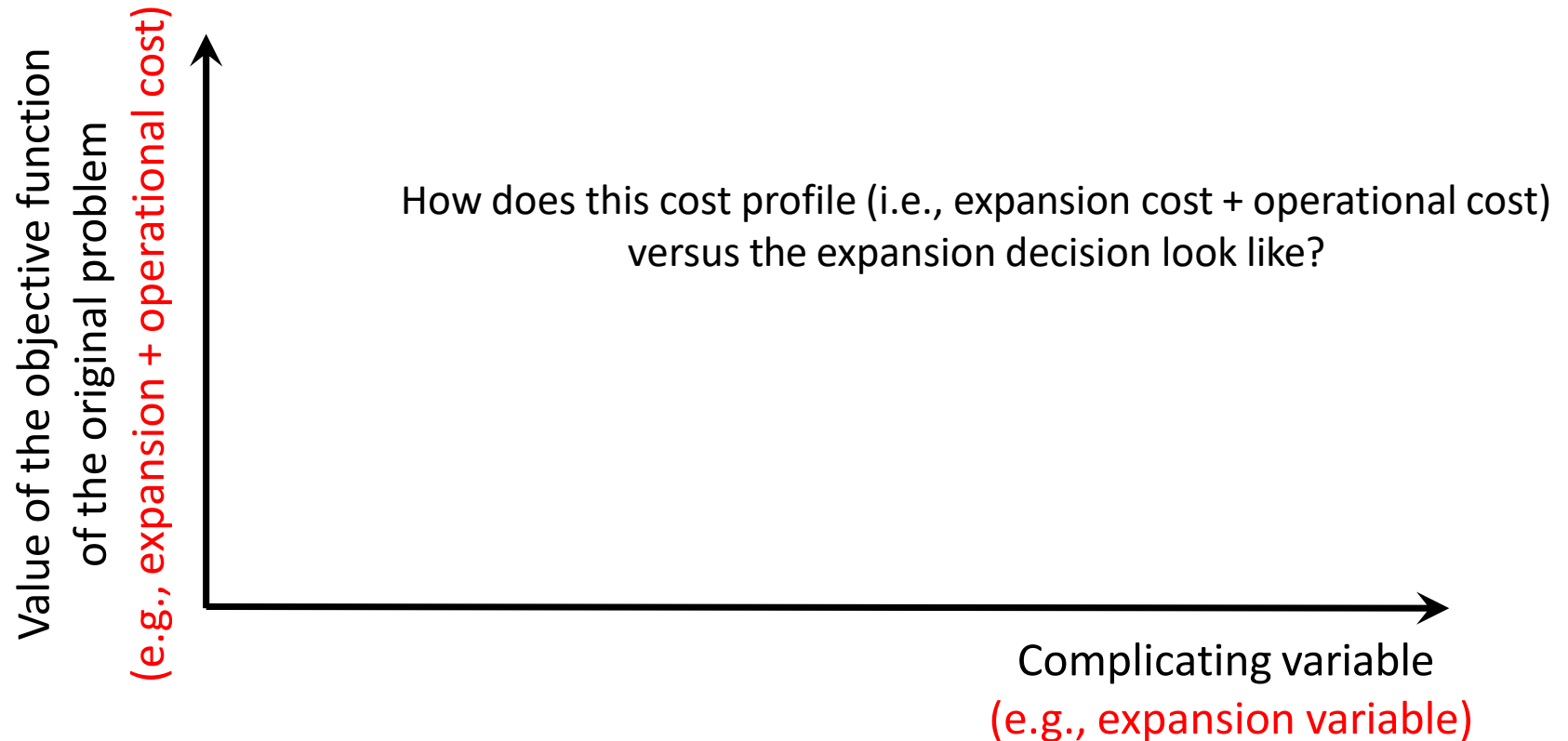
Jacobus Franciscus (Jacques) Benders

(1 June 1924 - 9 January 2017)

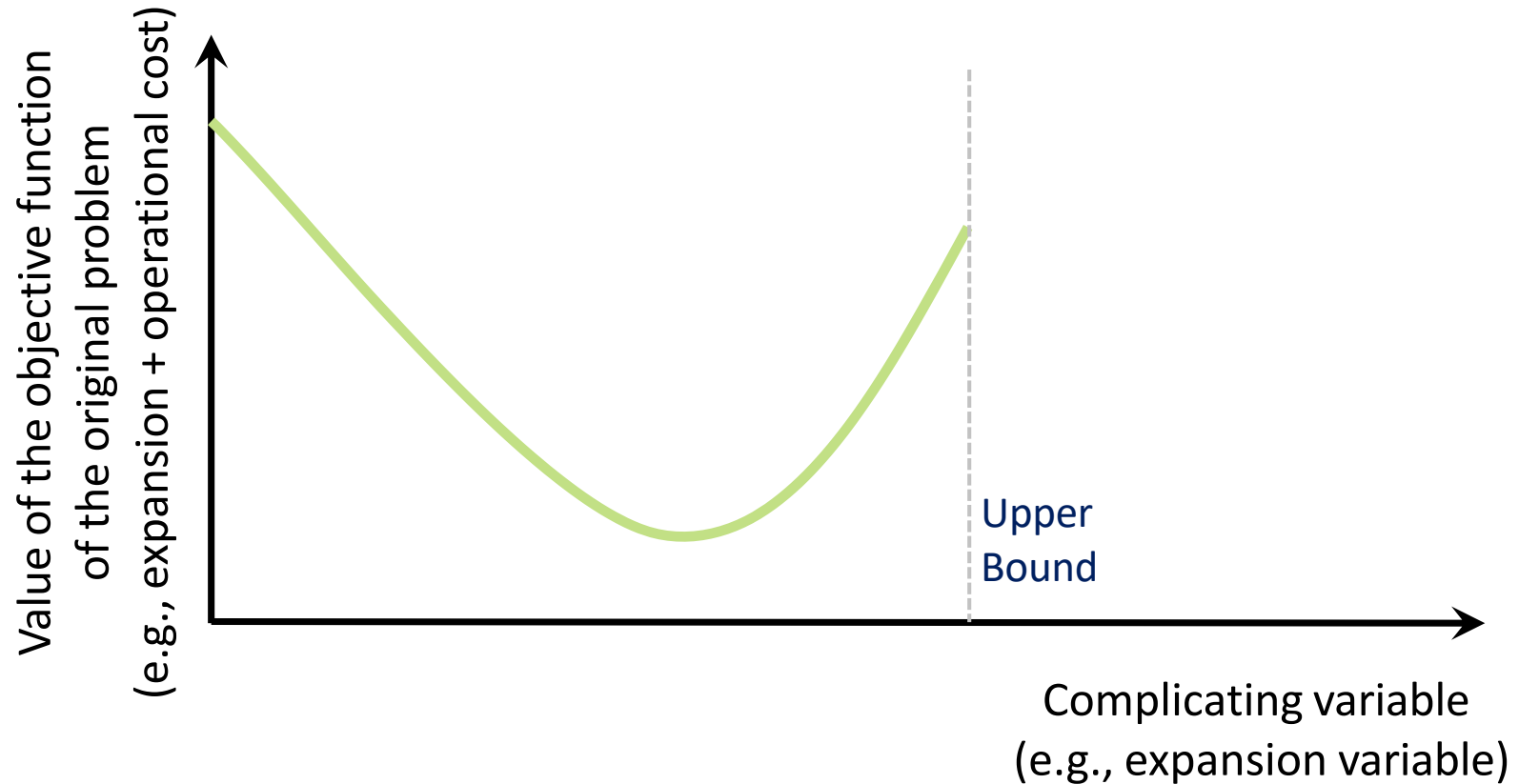
https://en.wikipedia.org/wiki/Jacques_F._Benders

J. F. Benders, "Partitioning procedures for solving mixed-variables programming problems," *Numerische Mathematik*, vol. 4, no. 1, pp. 238–252, 1962.

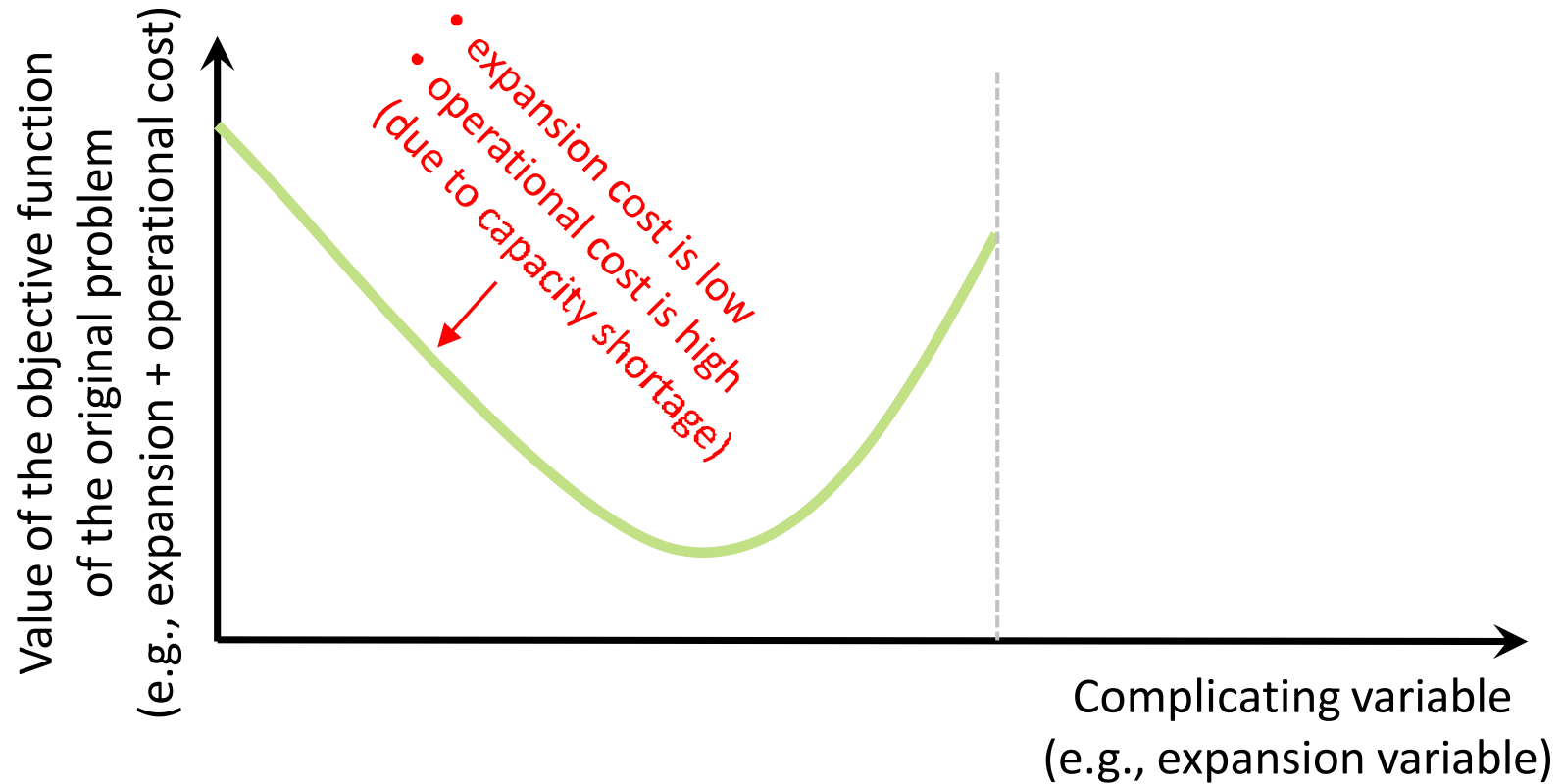
Concept-capacity expansion problem



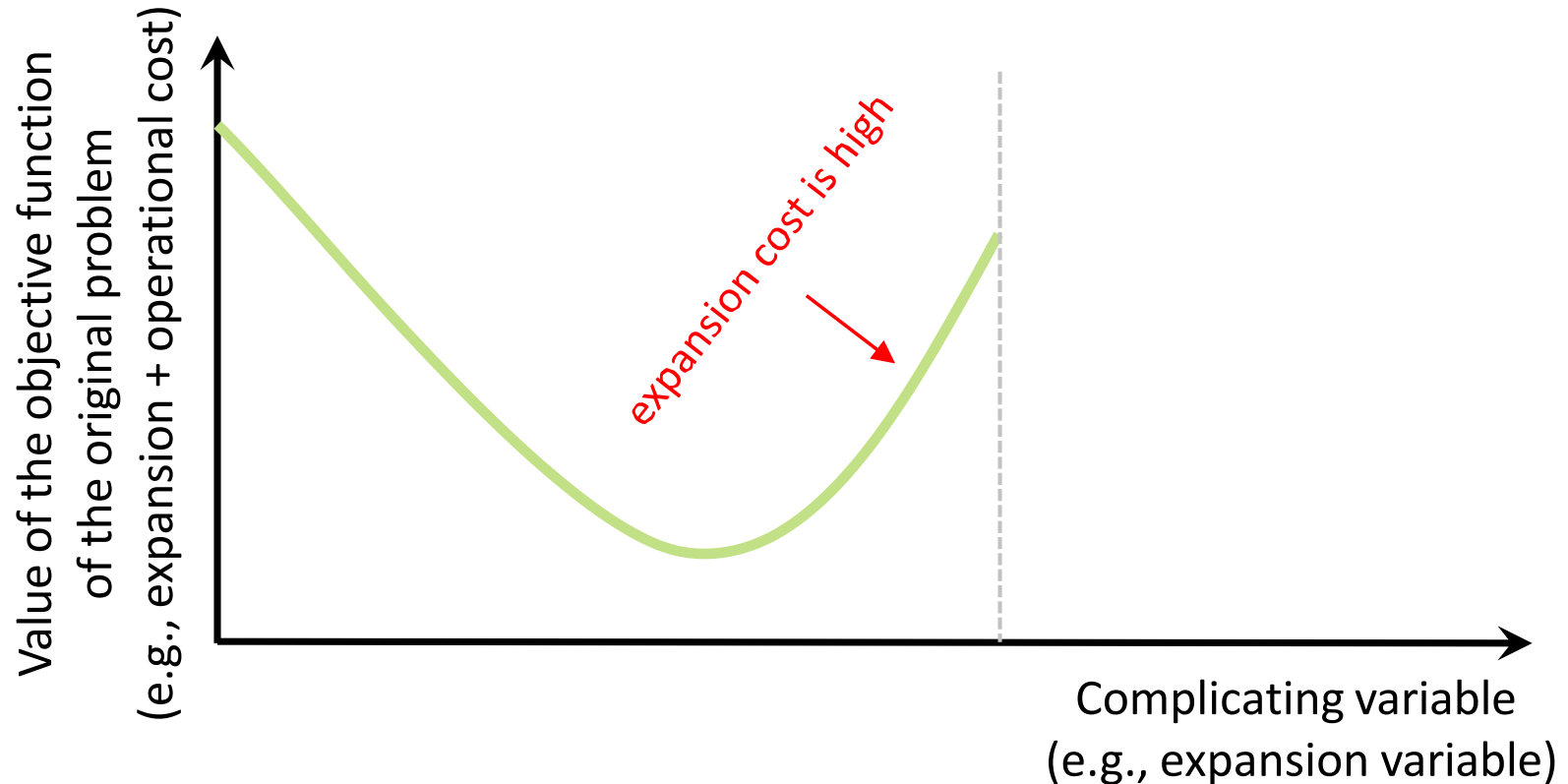
Concept-capacity expansion problem



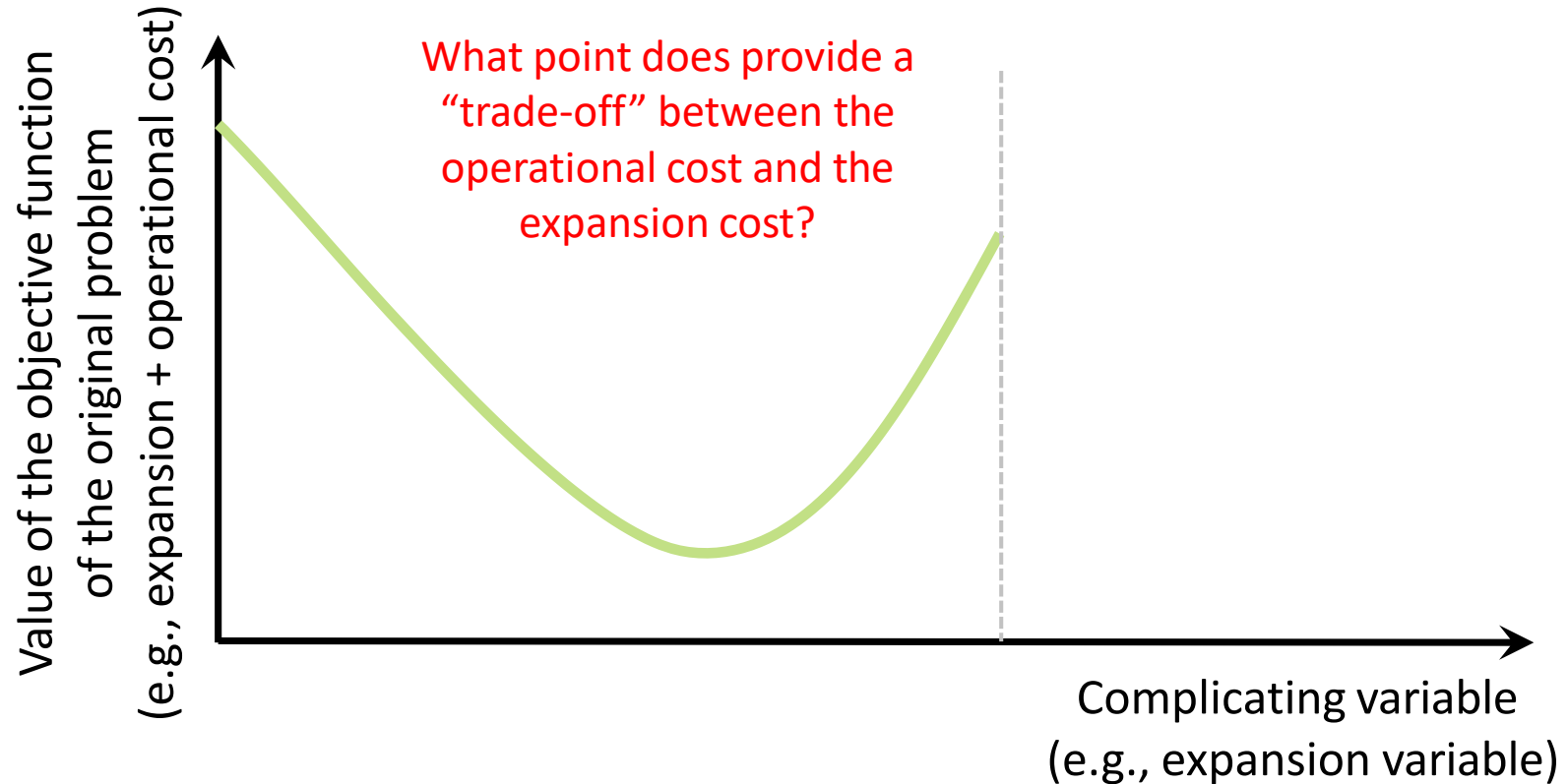
Concept-capacity expansion problem



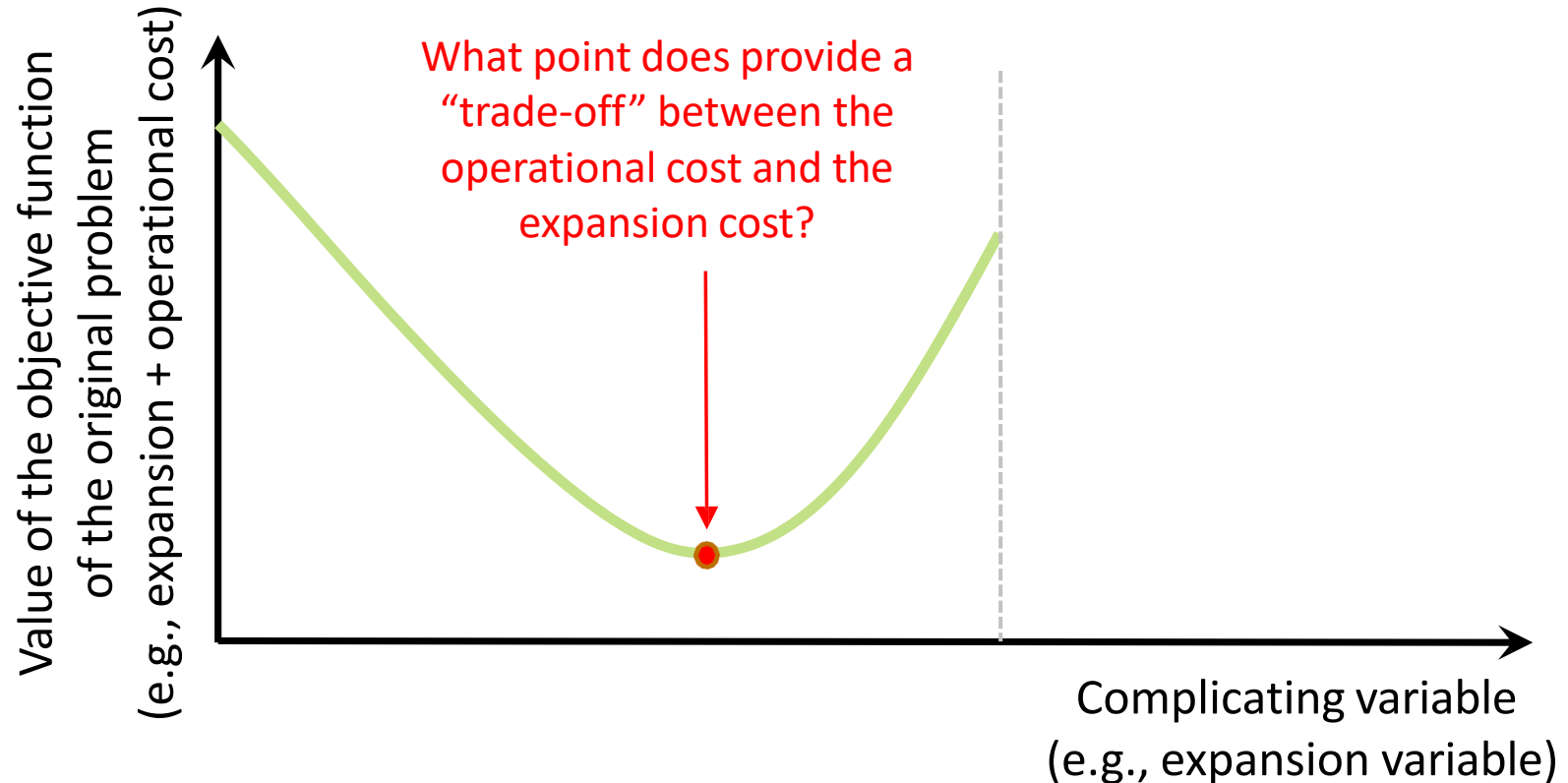
Concept-capacity expansion problem



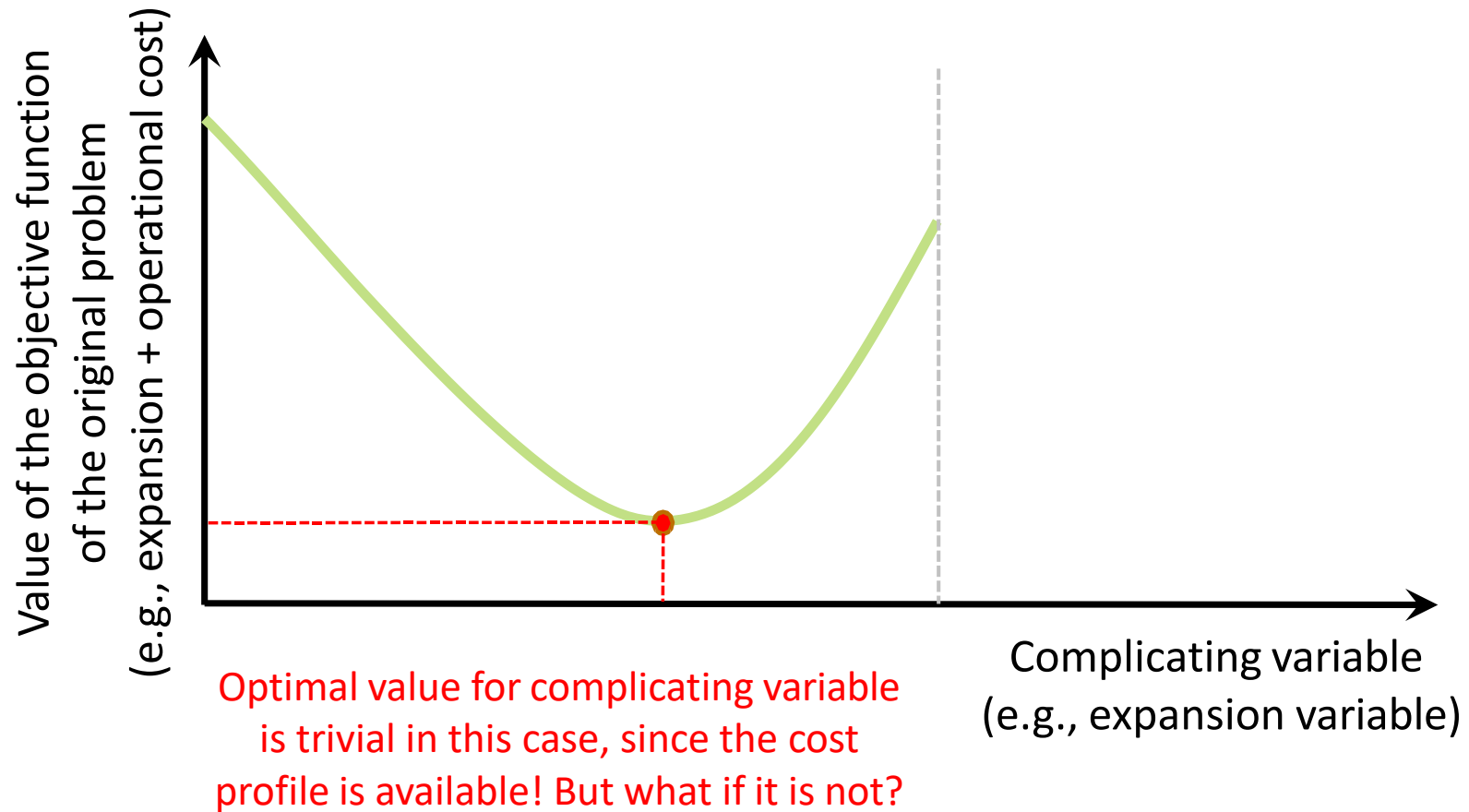
Concept-capacity expansion problem



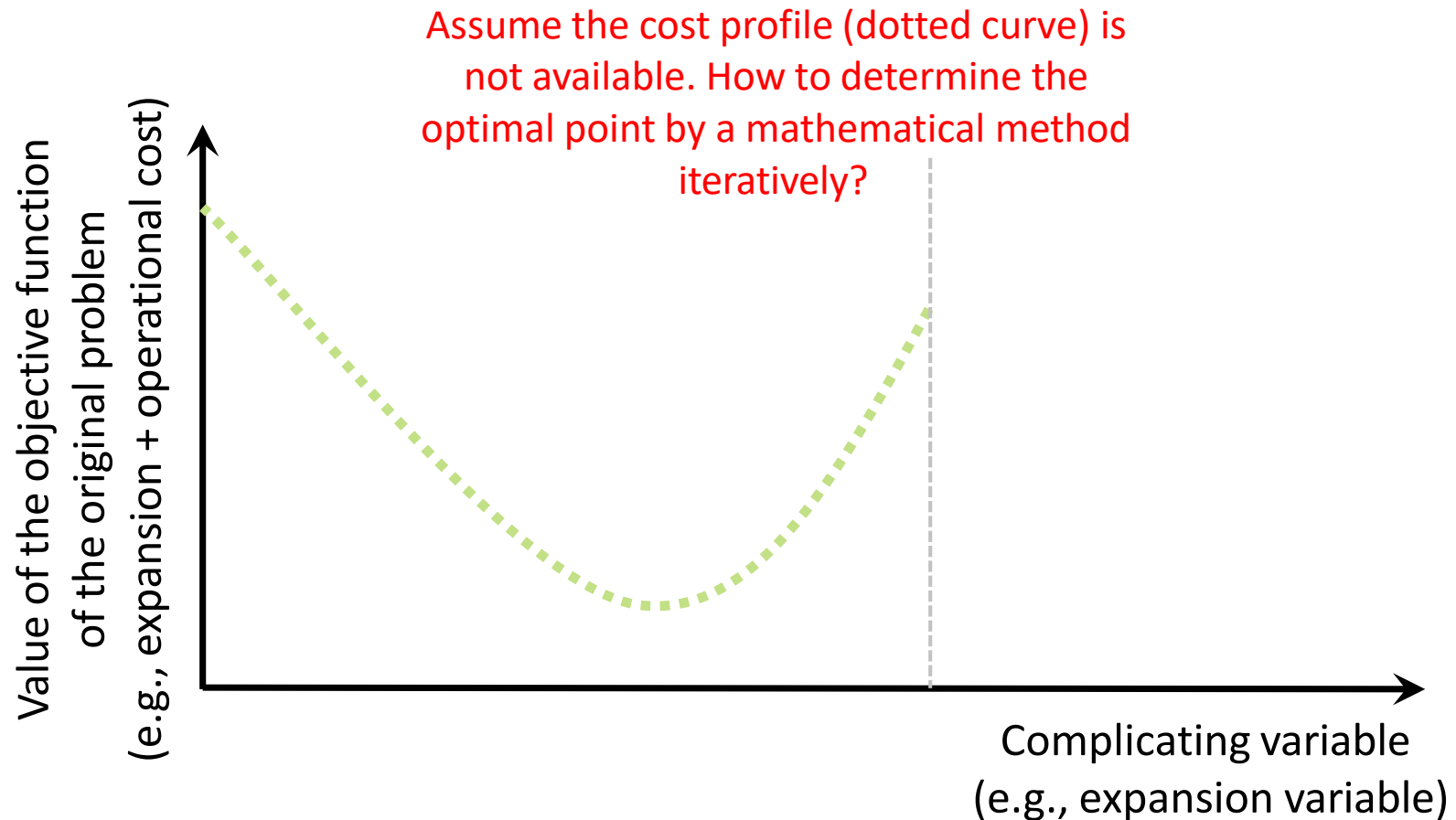
Concept-capacity expansion problem



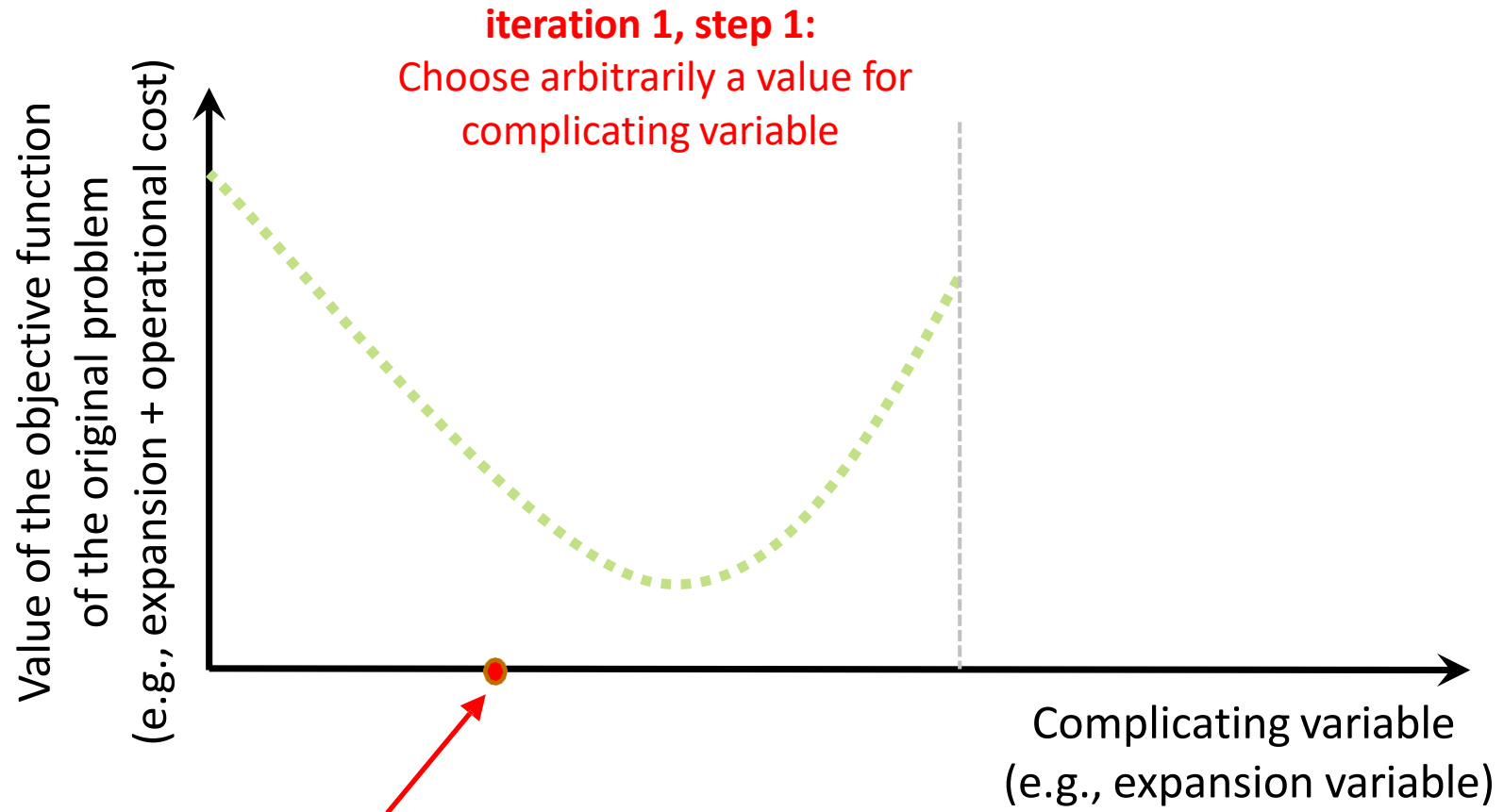
Concept-capacity expansion problem



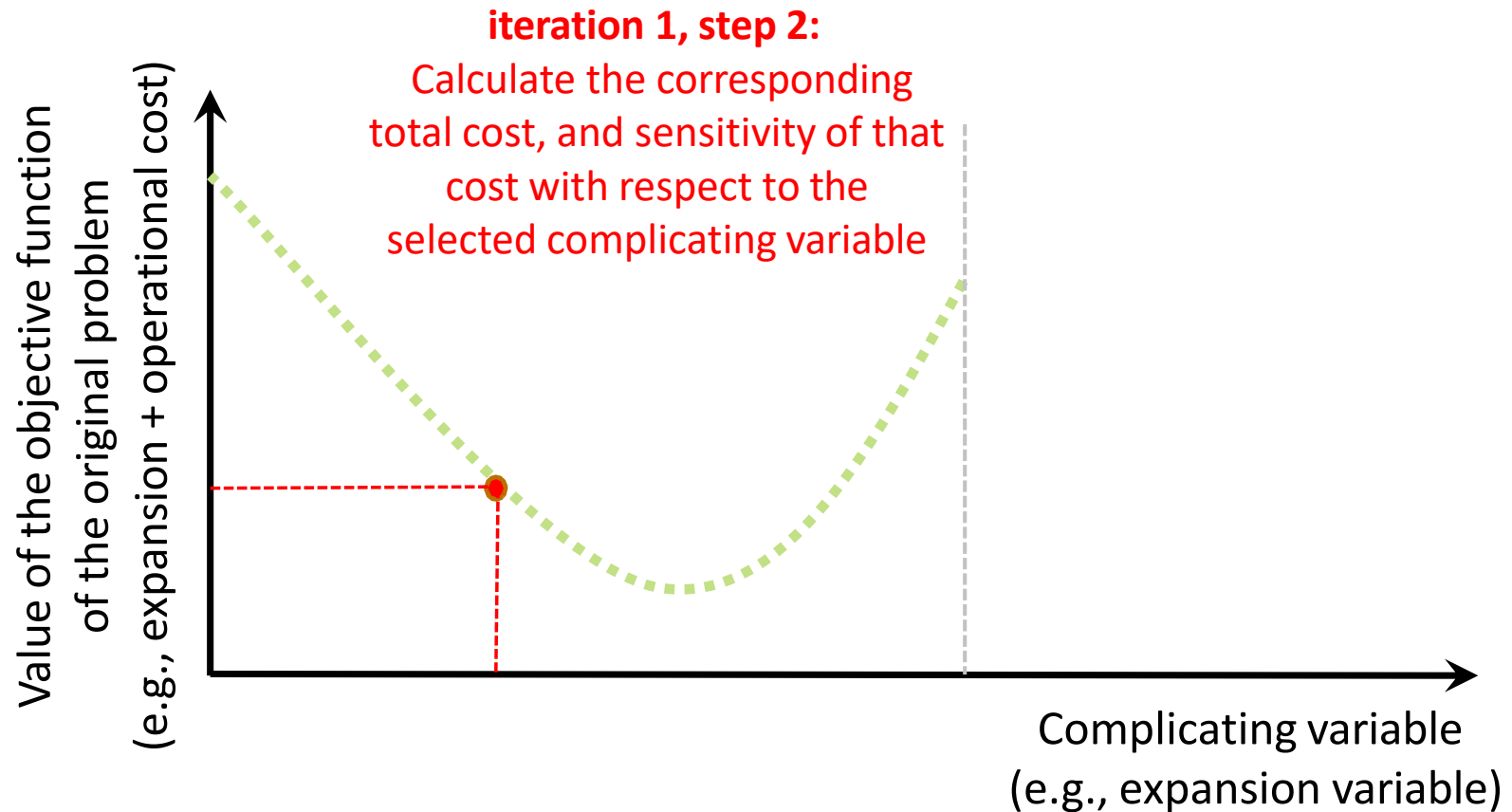
Concept-capacity expansion problem



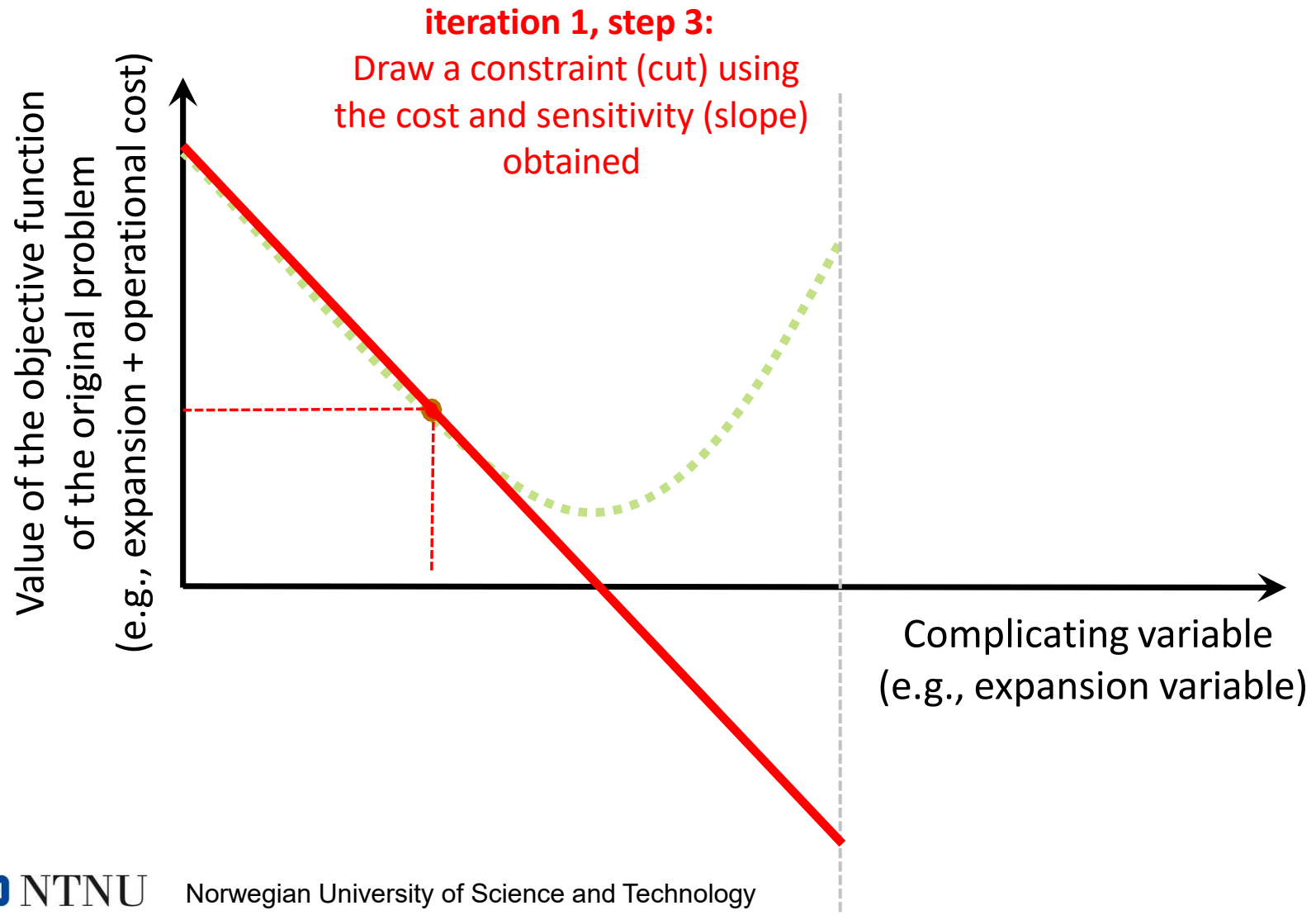
Concept-capacity expansion problem



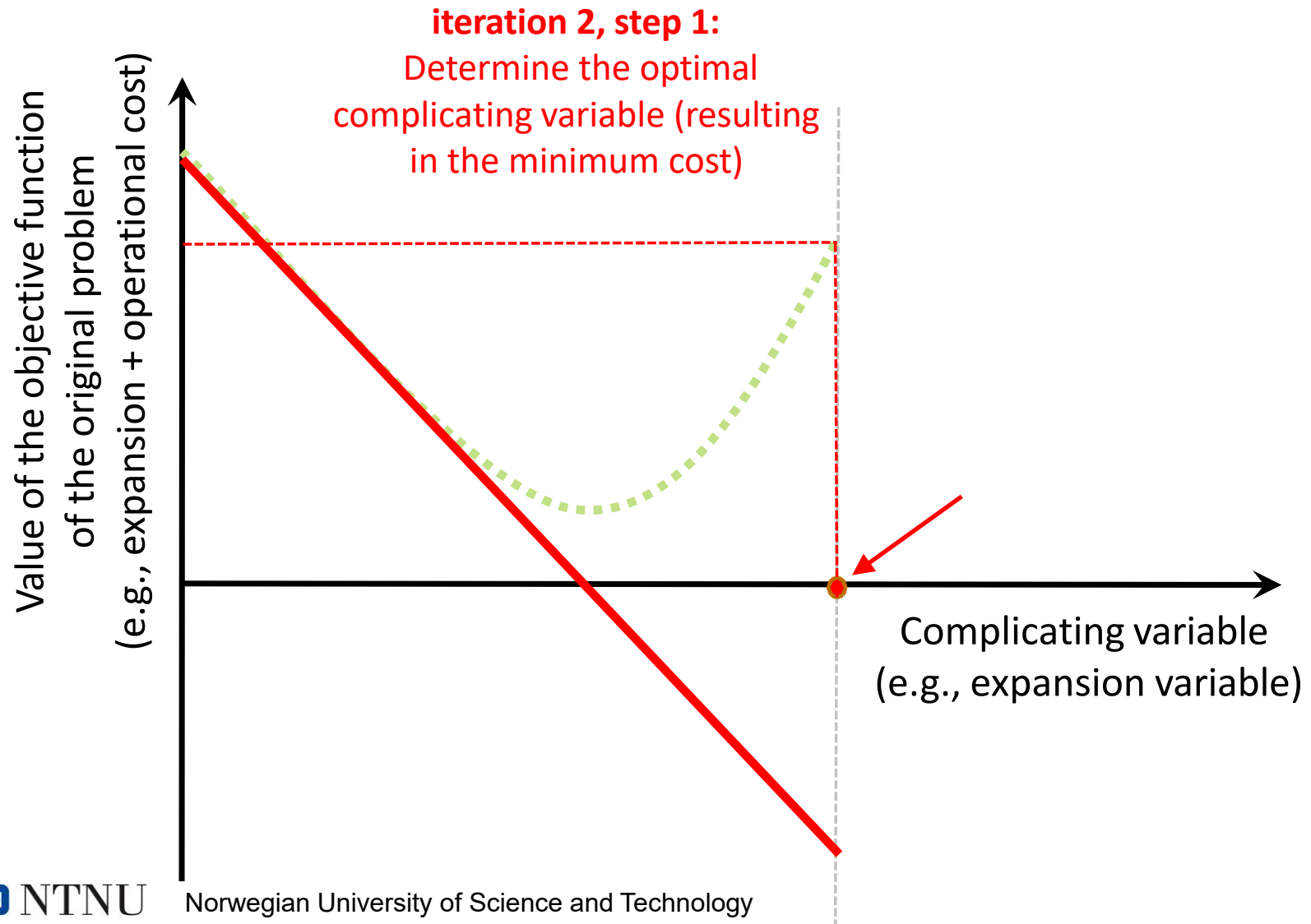
Concept-capacity expansion problem



Concept-capacity expansion problem

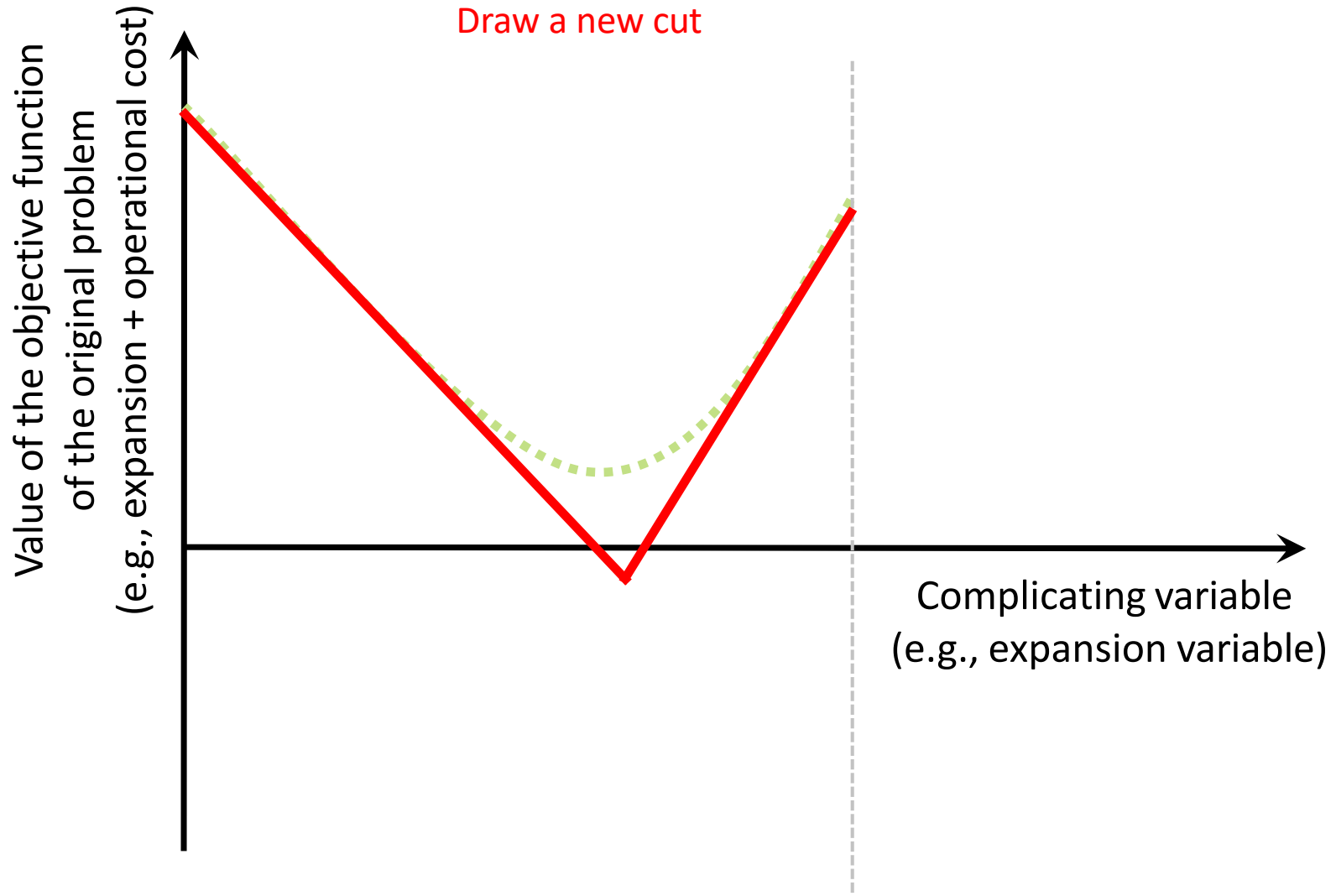


Concept-capacity expansion problem

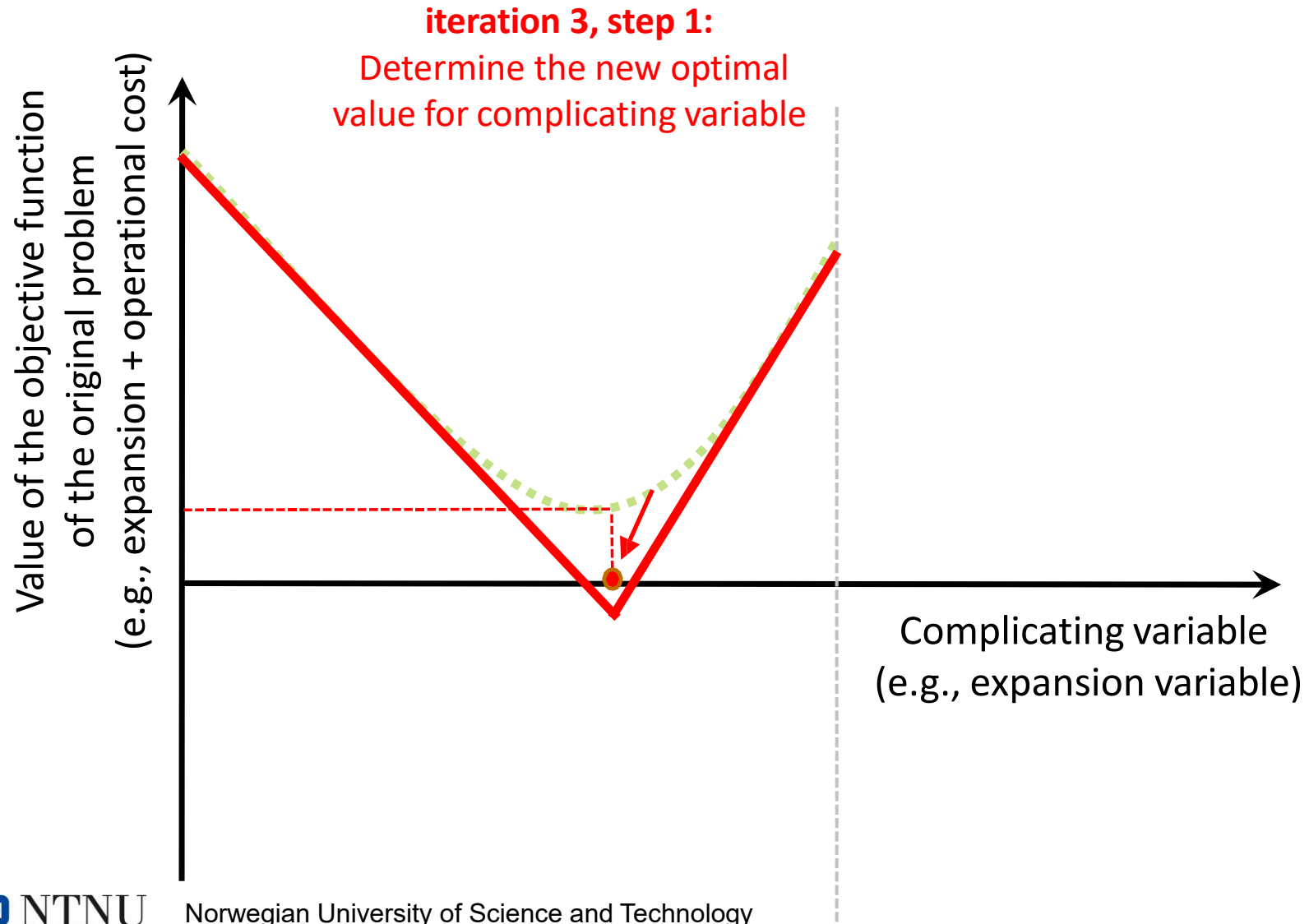


Concept-capacity expansion problem

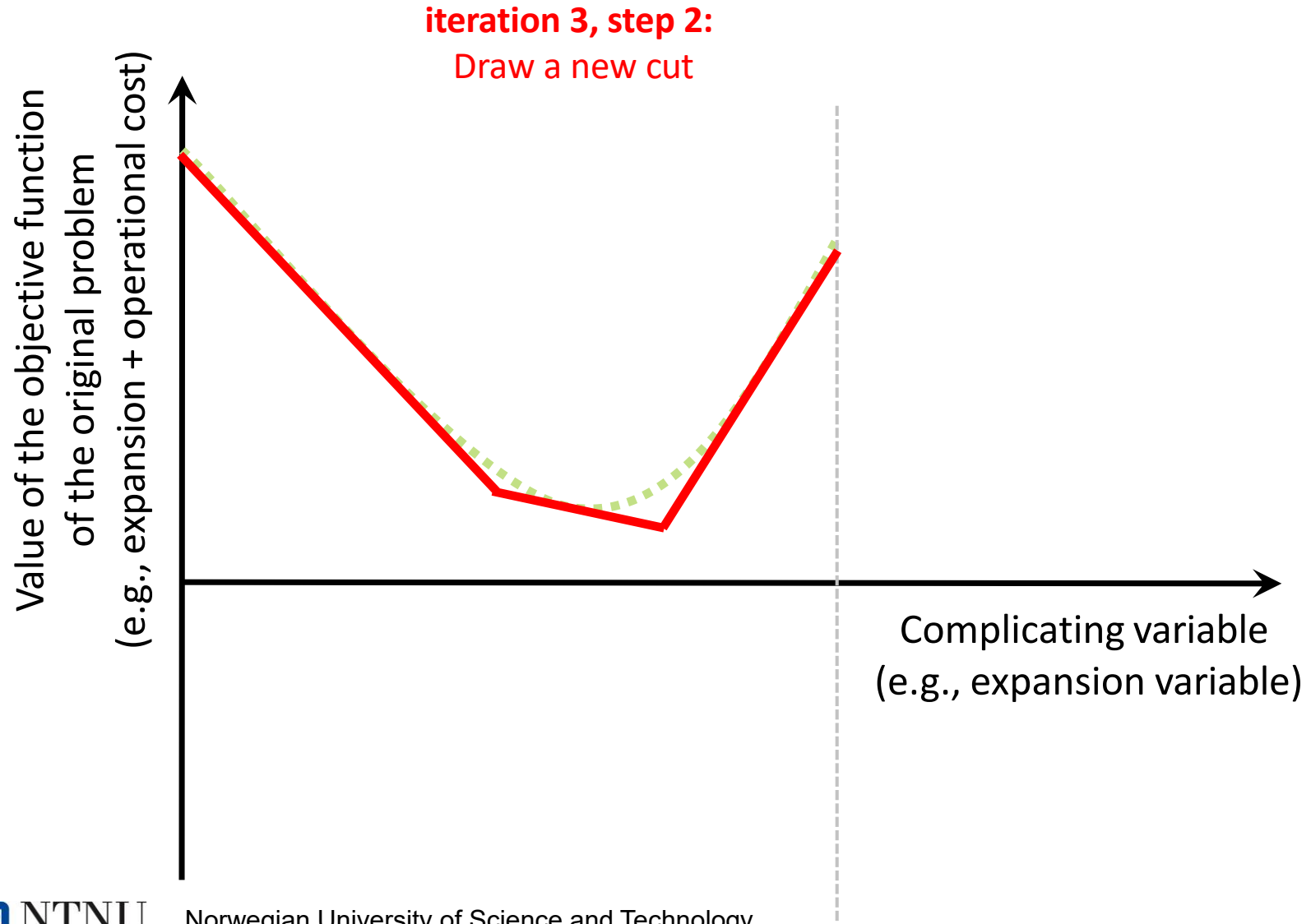
iteration 2, step 2:
Draw a new cut



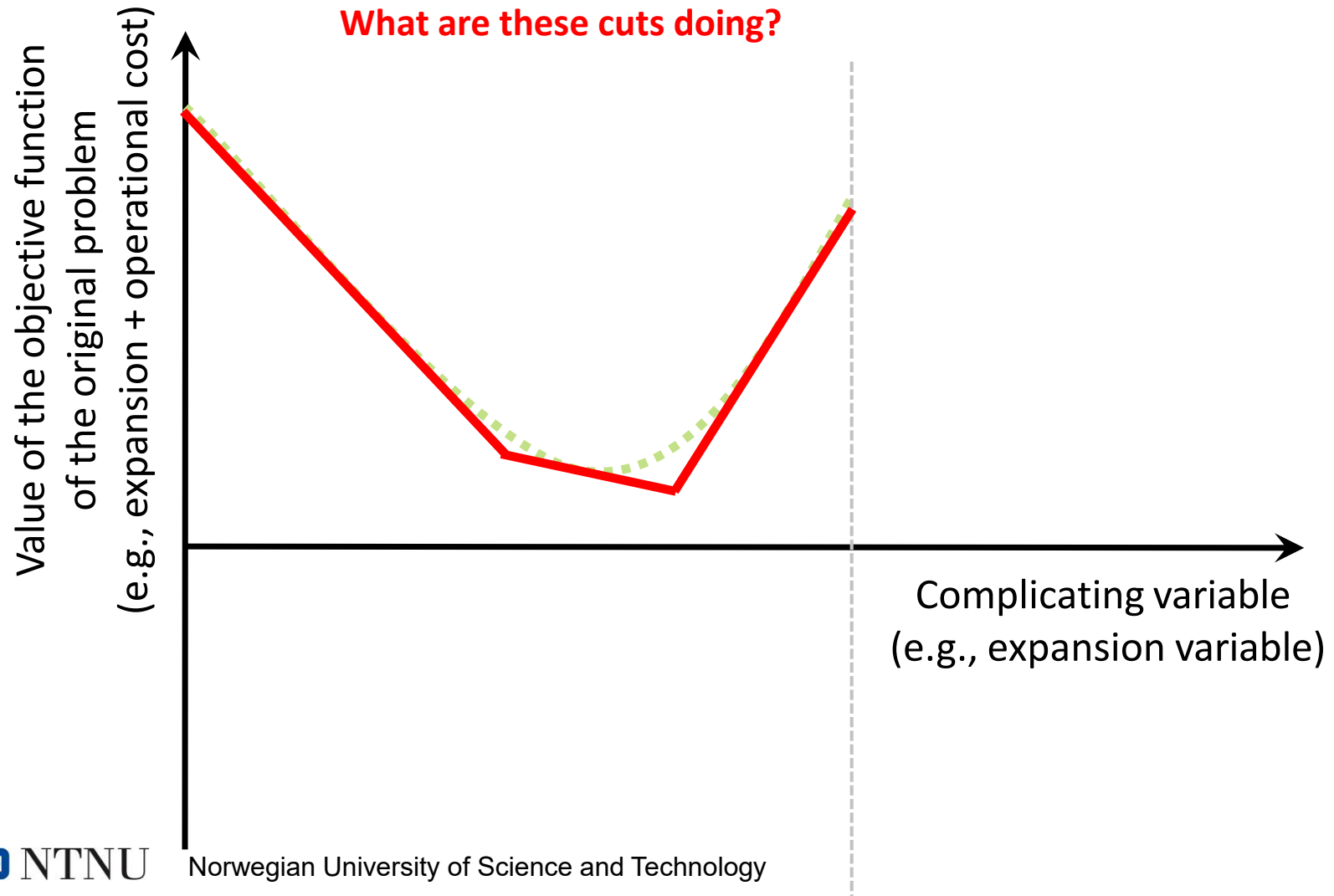
Concept-capacity expansion problem



Concept-capacity expansion problem

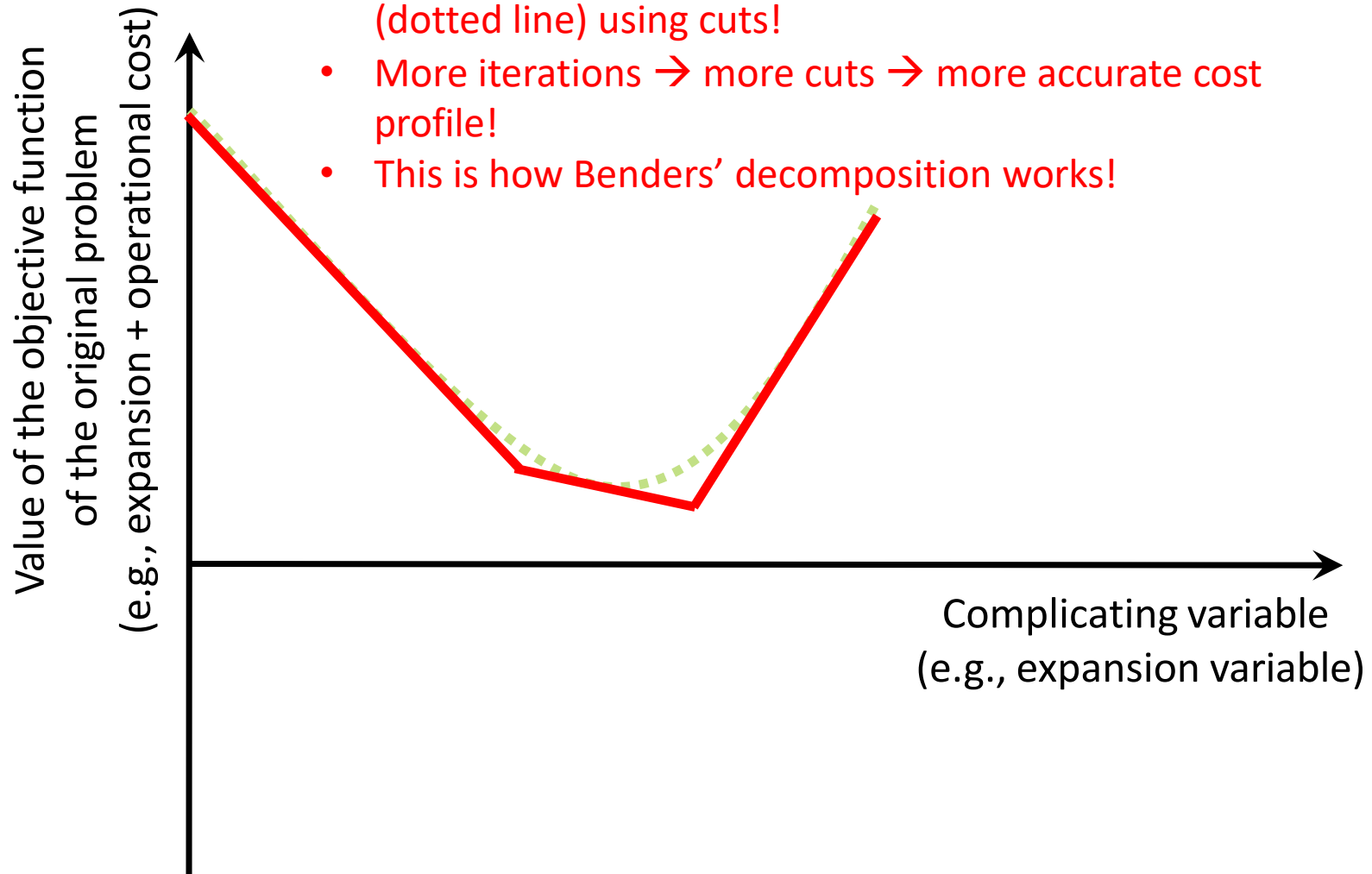


Concept-capacity expansion problem



Concept-capacity expansion problem

- We are indeed constructing the original cost profile (dotted line) using cuts!
- More iterations $\rightarrow$ more cuts $\rightarrow$ more accurate cost profile!
- This is how Benders' decomposition works!



Illustrative Example-Book section 3.3

$$\underset{x,y}{\text{minimise}} \quad z = -y - \frac{x}{4}$$

subject to,

$$y - x \leq 5$$

$$y - \frac{1}{2}x \leq \frac{15}{2}$$

$$y + \frac{1}{2}x \leq \frac{35}{2}$$

$$-y + x \leq 10$$

$$0 \leq x \leq 16$$

$$y \geq 0$$

Let's consider x as a complicating variable, and assume x is fixed!

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
- Based on values for sensitivities achieved in subproblem, master problem obtains the updated value for complicating variable x .

Subproblem:

$$\begin{aligned} & \underset{y^{(i)}}{\text{minimise}} -y^{(i)} \\ & \text{subject to,} \\ & y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)} \\ & y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)} \\ & y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)} \\ & -y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)} \\ & y^{(i)} \geq 0 \end{aligned}$$

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

subject to,

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\alpha^{(i)} \geq \pi^{(k)} [5 + x^{(i)}] + \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] + \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] + \gamma^{(k)} [10 - x^{(i)}] \quad \forall k = 1, \dots, i-1$$

This is an auxiliary variable, representing the objective function of the subproblem(s) within the objective function of the original problem!

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
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Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

subject to,

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

To avoid unbounded solution in the first iteration when there is no cut (last constraint). It should be a very large negative constant, e.g., -10^6 .

$$\alpha^{(i)} \geq \pi^{(k)} \left[5 + x^{(i)} \right] + \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] + \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] + \gamma^{(k)} \left[10 - x^{(i)} \right] \quad \forall k = 1, \dots, i-1$$

Illustrative Example

i : current Bender's iteration

k : set of previous iteration

- For given value for complicating variable (i.e., x), subproblem obtains the optimal value for y and dual variables (sensitivities).
- Based on values for sensitivities achieved in subproblem, master problem obtains the updated value for complicating variable x .

Master problem:

$$\underset{x^{(i)}, \alpha^{(i)}}{\text{minimise}} -\frac{x^{(i)}}{4} + \alpha^{(i)}$$

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$$\alpha^{(i)} \geq \pi^{(k)} \left[5 + x^{(i)} \right] + \mu^{(k)} \left[\frac{15}{2} + \frac{x^{(i)}}{2} \right] + \sigma^{(k)} \left[\frac{32}{2} - \frac{x^{(i)}}{2} \right] + \gamma^{(k)} \left[10 - x^{(i)} \right] \quad \forall k = 1, \dots, i-1$$

- The right hand side is the objective function of subproblem
- This constraint generates "Benders' (optimality) cuts". One additional cut in each iteration!

Subproblem:

$$\underset{y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} \leq 5 + x^{\text{fixed}(i)} \quad : \pi^{(i)}$$

$$y^{(i)} \leq \frac{15}{2} + \frac{x^{\text{fixed}(i)}}{2} \quad : \mu^{(i)}$$

$$y^{(i)} \leq \frac{32}{2} - \frac{x^{\text{fixed}(i)}}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} \leq 10 - x^{\text{fixed}(i)} \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

Algorithm

- **Step 0: Initialization**

Set $i = 1$, $x^{\text{fixed}(1)} = x^{\text{initial}}$, and lower bound (LB) $= -\infty$

- **Step 1: Solve subproblem(s):** obtain the values of all dual variables (sensitivities), and the value of objective function, which is upper bound (UB)

- **Step 2: Convergence check**

if $|UB - LB| \leq \epsilon$, then the optimal solution with a level of accuracy ϵ is obtained, otherwise $i \leftarrow i + 1$

- **Step 3: Solve master problem**

obtain the updated $x^{(i)}$ and the value of $\alpha^{(i)}$ as the updated LB, and go Step 1 with the updated fixed x

A compact form

Subproblem:

$$\underset{x^{(i)}, y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} - x^{(i)} \leq 5 \quad : \pi^{(i)}$$

$$y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} \quad : \mu^{(i)}$$

$$y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} + x^{(i)} \leq 10 \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

$$x^{(i)} = x^{\text{fixed}(i)} \quad : \rho^{(i)}$$

A compact form

In this compact form, there is a constraint for fixing each complicating variable to a given value! However, x is now a “variable” in subproblem, though its value is fixed! Only one sensitivity ($\rho^{(i)}$) is needed in the master problem!

Subproblem:

$$\underset{x^{(i)}, y^{(i)}}{\text{minimise}} -y^{(i)}$$

subject to,

$$y^{(i)} - x^{(i)} \leq 5 \quad : \pi^{(i)}$$

$$y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} \quad : \mu^{(i)}$$

$$y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} \quad : \sigma^{(i)}$$

$$-y^{(i)} + x^{(i)} \leq 10 \quad : \gamma^{(i)}$$

$$y^{(i)} \geq 0$$

$$x^{(i)} = x^{\text{fixed}(i)} \quad : \rho^{(i)}$$

A compact form

Subproblem:

$$\begin{aligned} & \underset{x^{(i)}, y^{(i)}}{\text{minimise}} && -y^{(i)} \\ & \text{subject to,} && \\ & y^{(i)} - x^{(i)} \leq 5 && : \pi^{(i)} \\ & y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} && : \mu^{(i)} \\ & y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} && : \sigma^{(i)} \\ & -y^{(i)} + x^{(i)} \leq 10 && : \gamma^{(i)} \\ & y^{(i)} \geq 0 && \\ & x^{(i)} = x^{\text{fixed}(i)} && : \rho^{(i)} \end{aligned}$$

Master problem:

$$\begin{aligned} & \underset{x^{(i)}, \alpha^{(i)}}{\text{minimize}} && -\frac{x^{(i)}}{4} + \alpha^{(i)} \\ & \text{subject to} && \\ & 0 \leq x^{(i)} \leq 16 && \\ & \alpha^{(i)} \geq \alpha^{\text{down}} && \\ & \alpha^{(i)} \geq -y^{(k)} + \rho^{(k)}[x^{(i)} - x^{(k)}] \quad \forall k = 1, \dots, i-1 && \end{aligned}$$

A compact form

Subproblem:

$$\begin{aligned} & \underset{x^{(i)}, y^{(i)}}{\text{minimise}} && -y^{(i)} \\ & \text{subject to,} && \\ & y^{(i)} - x^{(i)} \leq 5 && : \pi^{(i)} \\ & y^{(i)} - \frac{x^{(i)}}{2} \leq \frac{15}{2} && : \mu^{(i)} \\ & y^{(i)} + \frac{x^{(i)}}{2} \leq \frac{32}{2} && : \sigma^{(i)} \\ & -y^{(i)} + x^{(i)} \leq 10 && : \gamma^{(i)} \\ & y^{(i)} \geq 0 && \\ & x^{(i)} = x^{\text{fixed}(i)} && : \rho^{(i)} \end{aligned}$$

Master problem:

$$\begin{aligned} & \underset{x^{(i)}, \alpha^{(i)}}{\text{minimize}} && -\frac{x^{(i)}}{4} + \alpha^{(i)} \\ & \text{subject to} && \end{aligned}$$

$$0 \leq x^{(i)} \leq 16$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\alpha^{(i)} \geq -y^{(k)} + \rho^{(k)}[x^{(i)} - x^{(k)}] \quad \forall k = 1, \dots, i-1$$

A compact formulation for
Benders' cut!



Applications of Benders' decomposition to two-stage scenario-based stochastic programming

Two-stage scenario-based stochastic programming

First stage cost



$$\text{Minimize}_{x, y_\omega} \boxed{f(x)} + \sum_{\omega} \pi_{\omega} g(y_{\omega})$$

subject to

$$h(x) \leq a$$

$$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$$

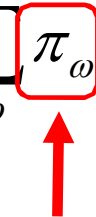
Two-stage scenario-based stochastic programming

Minimize $f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$

subject to

$h(x) \leq a$

$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$

 Probability of scenario w

Two-stage scenario-based stochastic programming

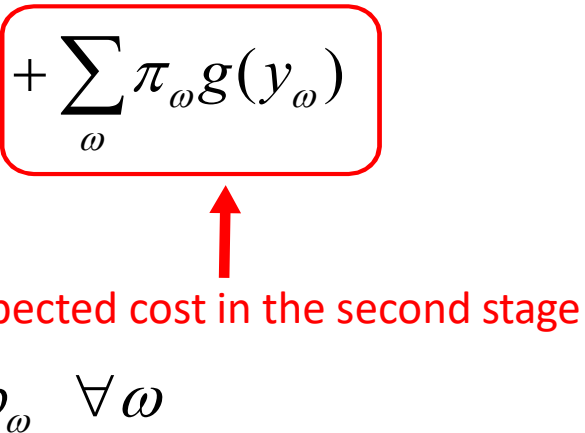
Minimize $f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$

subject to

$h(x) \leq a$

$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$

Expected cost in the second stage



Two-stage scenario-based stochastic programming

$$\underset{x, y_\omega}{\text{Minimize}} \quad f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega})$$

subject to

$$h(x) \leq a$$



First-stage constraint

$$k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega$$



Linking constraint

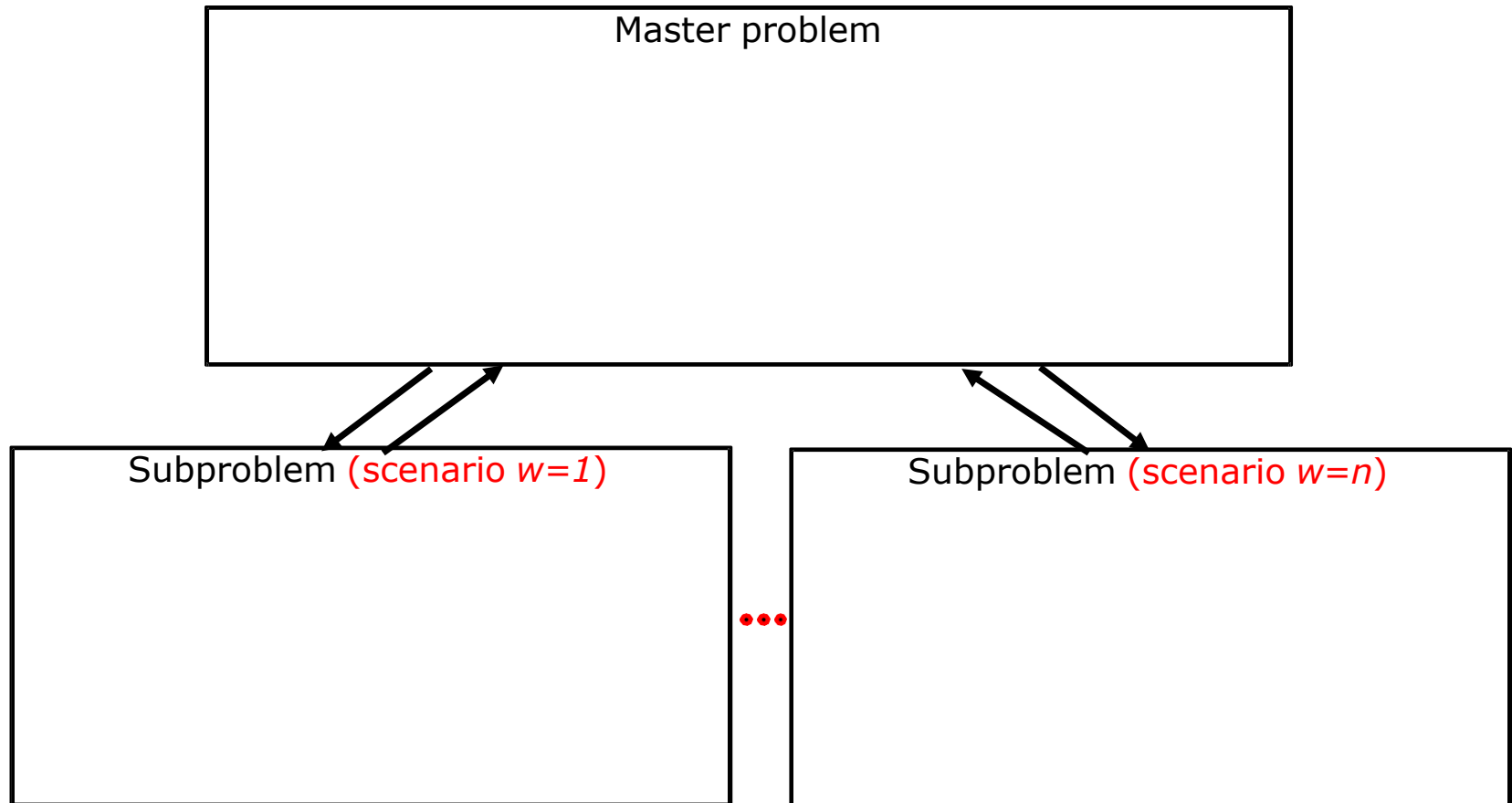
Two-stage scenario-based stochastic programming

$$\begin{aligned} & \underset{x, y_\omega}{\text{Minimize}} && f(x) + \sum_{\omega} \pi_{\omega} g(y_{\omega}) \\ & \text{subject to} && \\ & && h(x) \leq a \\ & && k(x) + l(y_{\omega}) \leq b_{\omega} \quad \forall \omega \end{aligned}$$

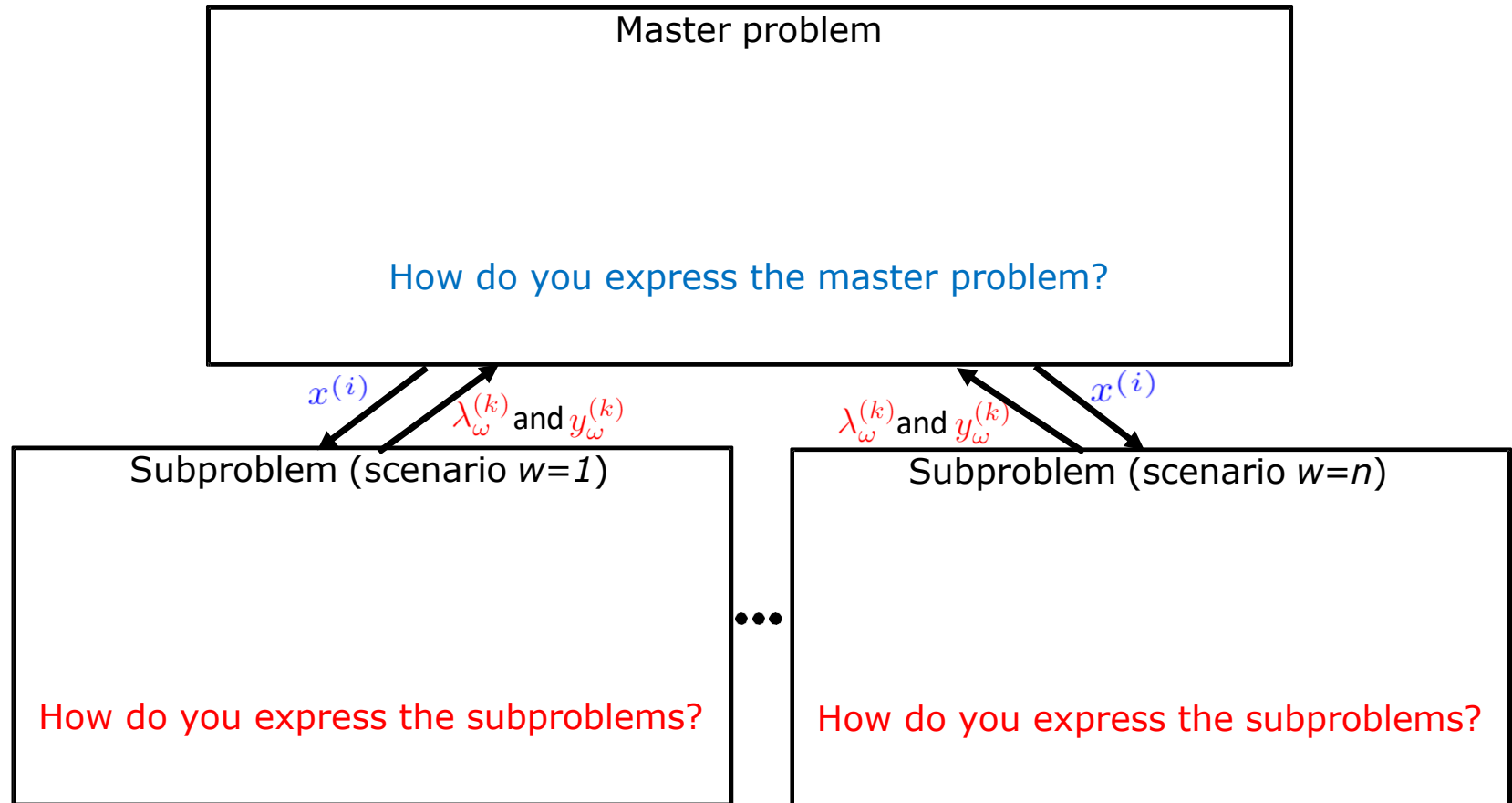
Recall that the first-stage variables (vector $\mathbf{x}$) are complicating variables!

- We need to formulate:
 - One master problem
 - One subproblem per scenario

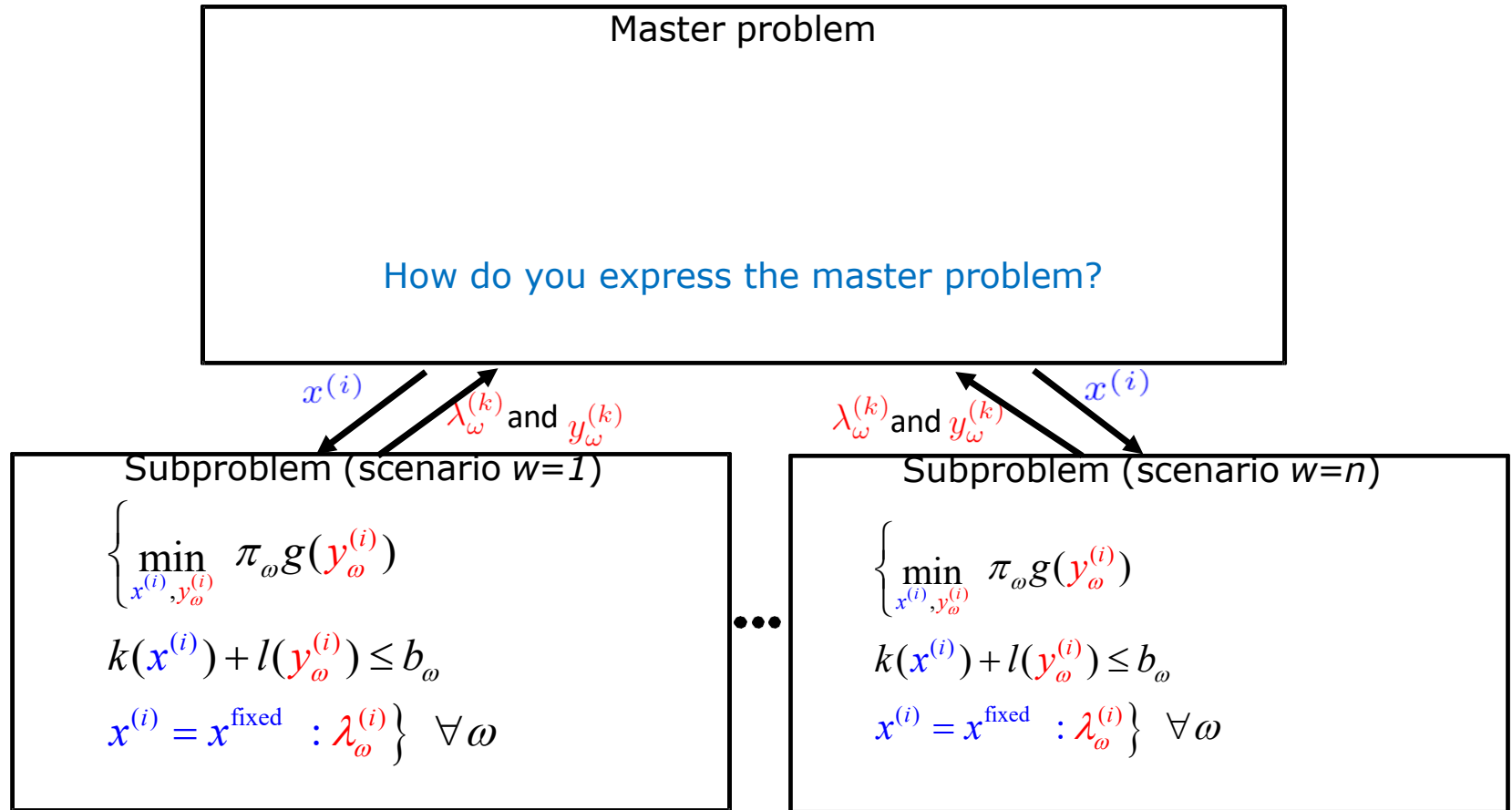
Two-stage stochastic program: Benders' solution



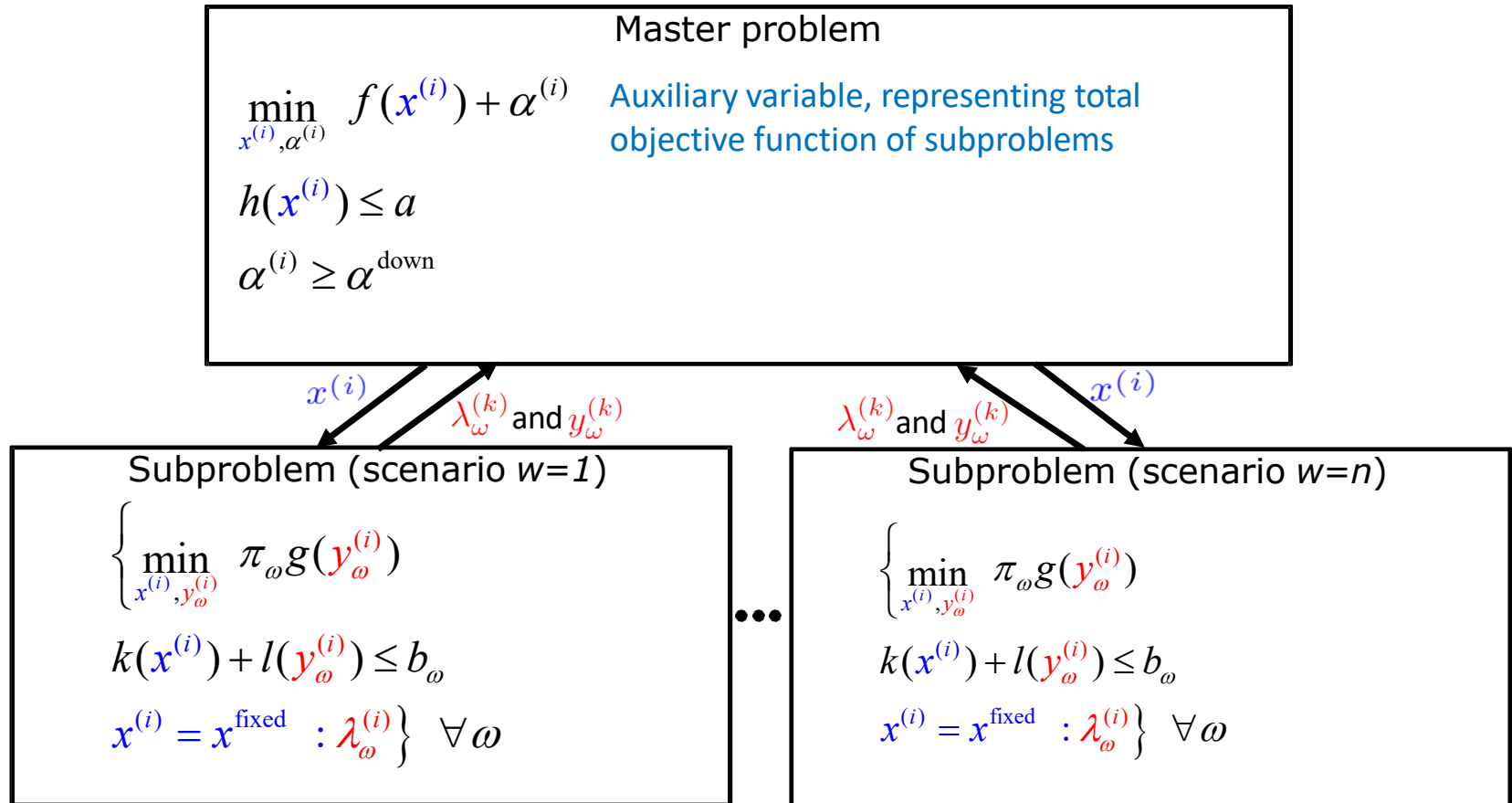
Two-stage stochastic program: Benders' solution



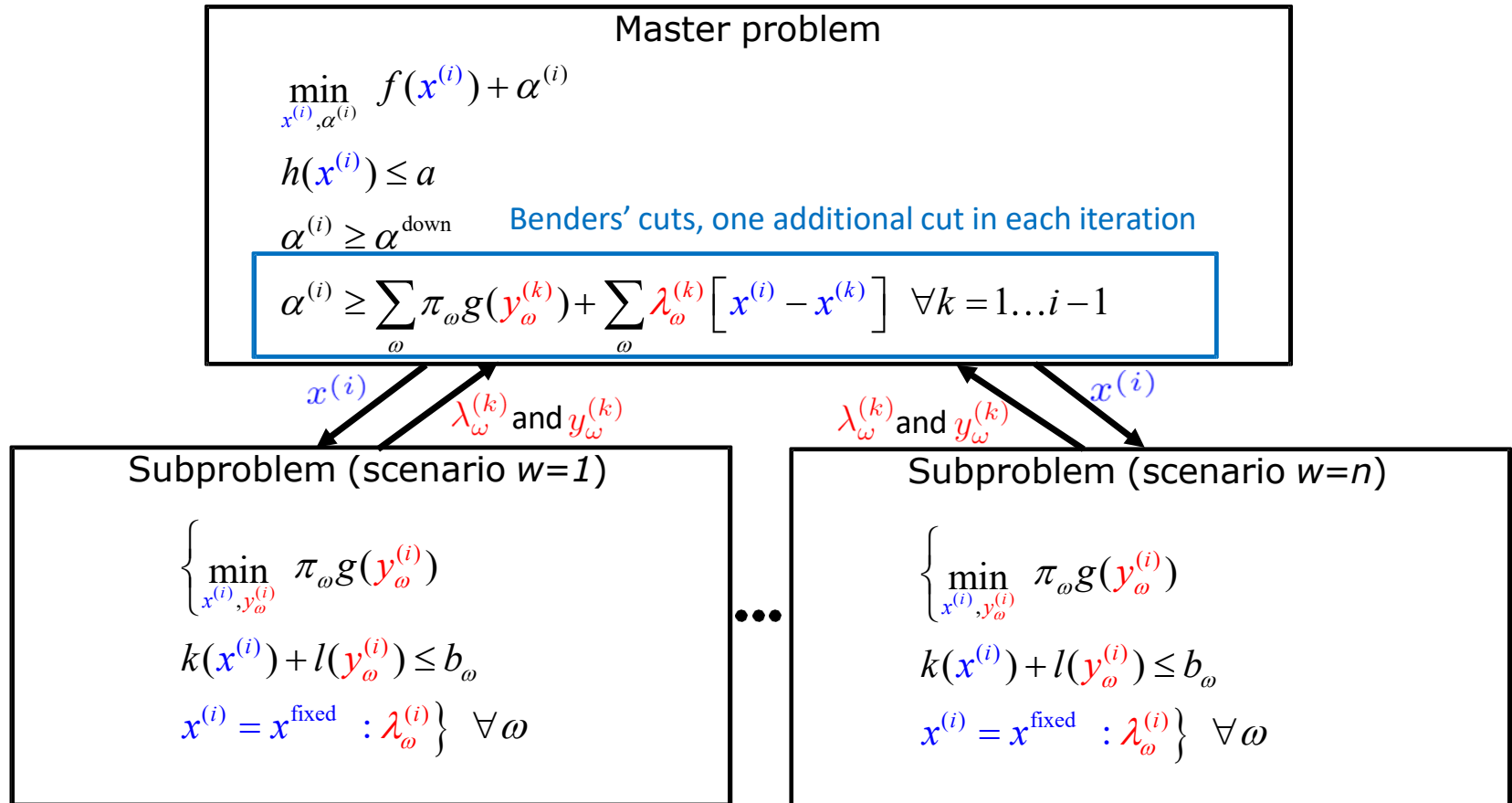
Two-stage stochastic program: Benders' solution



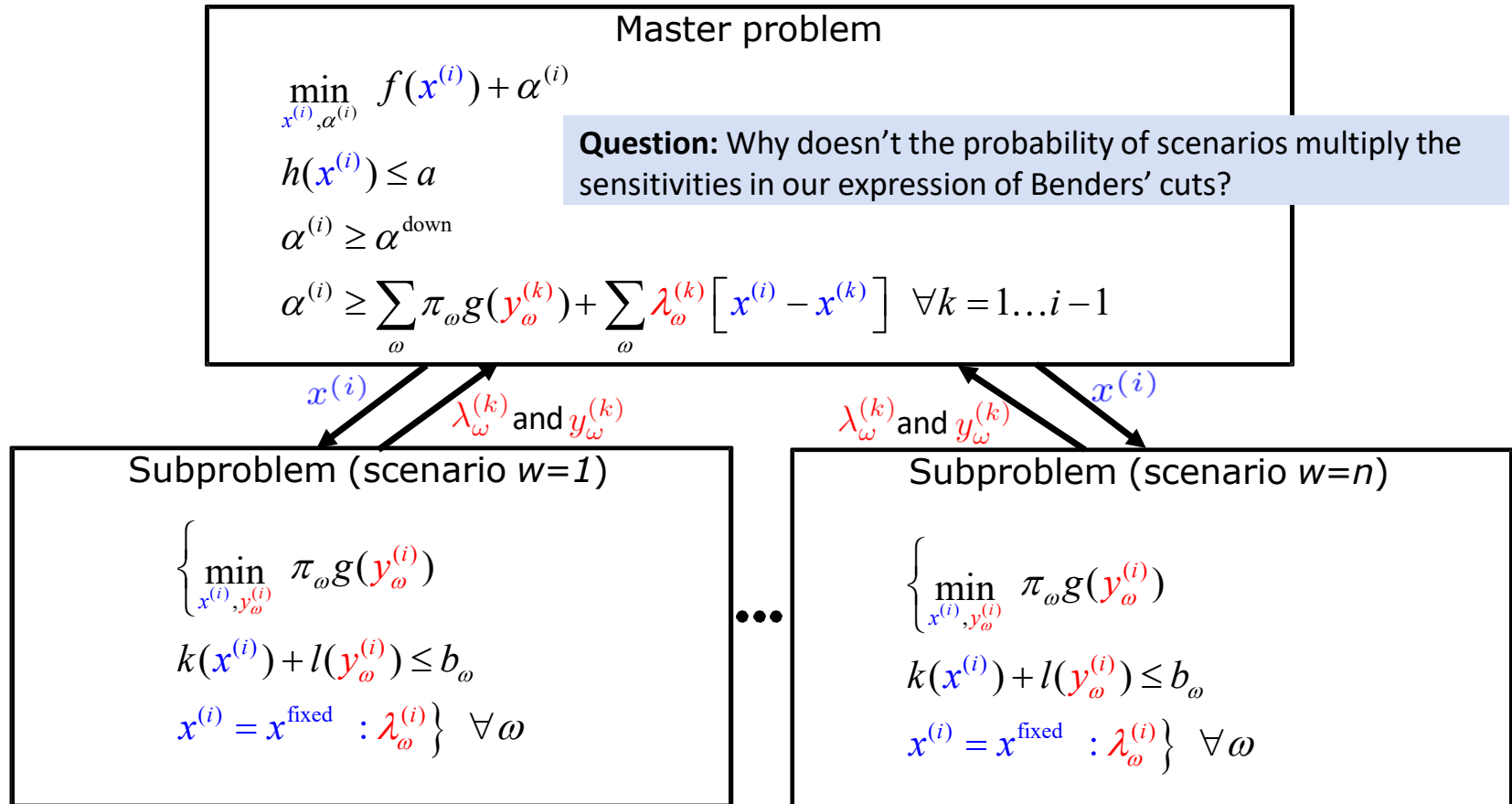
Two-stage stochastic program: Benders' solution



Two-stage stochastic program: Benders' solution



Two-stage stochastic program: Benders' solution



Decomposed model

Master problem

$$\begin{aligned}
 \min_{x^{(i)}, \alpha^{(i)}} & f(x^{(i)}) + \alpha^{(i)} \\
 & h(x^{(i)}) \leq a \\
 & \alpha^{(i)} \geq \alpha^{\text{down}} \\
 & \alpha^{(i)} \geq \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(k)}) \\
 & \quad + \sum_{\omega} \lambda_{\omega}^{(k)} [x^{(i)} - x^{(k)}] \\
 & \forall k = 1 \dots i-1
 \end{aligned}$$

Subproblem

$$\begin{aligned}
 & \left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} \pi_{\omega} g(y_{\omega}^{(i)}) \right. \\
 & k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega} \\
 & \left. x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \right\} \quad \forall \omega
 \end{aligned}$$

Decomposed model: Equivalent form

Master problem

$$\min_{x^{(i)}, \alpha^{(i)}} f(x^{(i)}) + \alpha^{(i)}$$

$$h(x^{(i)}) \leq a$$

$$\alpha^{(i)} \geq \alpha^{\text{down}}$$

$$\alpha^{(i)} \geq \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(k)})$$

$$+ \sum_{\omega} \pi_{\omega} \lambda_{\omega}^{(k)} [x^{(i)} - x^{(k)}]$$

$$\forall k = 1 \dots i-1$$

Subproblem

$$\left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} g(y_{\omega}^{(i)}) \right.$$

$$k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega}$$

$$x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \} \quad \forall \omega$$

Multi-cut Benders Decomposition

Concept

- Extension of classical Benders decomposition.
- Instead of one aggregated cut per iteration → one cut per subproblem.
- Useful in stochastic programming (one subproblem per scenario).

Master problem

$$\min_{x^{(i)}, \alpha^{(i)}} f(x^{(i)}) + \sum_{\omega} \pi_{\omega} \alpha_{\omega}$$

$$h(x^{(i)}) \leq a$$

$$\alpha_{\omega}^{(i)} \geq \alpha_{\omega}^{\text{down}} \quad \forall \omega \in \Omega$$

$$\alpha_{\omega}^{(i)} \geq g_{\omega}(y_{\omega}^{(k)}) + \lambda_{\omega}^{(k)} [x^{(i)} - x_{\omega}^{(k)}]$$

$$\forall k = 1 \dots i-1, \forall \omega \in \Omega$$

Subproblem

$$\left\{ \min_{x^{(i)}, y_{\omega}^{(i)}} g(y_{\omega}^{(i)}) \right.$$

$$k(x^{(i)}) + l(y_{\omega}^{(i)}) \leq b_{\omega}$$

$$x^{(i)} = x^{\text{fixed}} : \lambda_{\omega}^{(i)} \left. \right\} \quad \forall \omega \in \Omega$$

Multi-cut Benders Decomposition

Advantages

- Provides richer information to the master problem.
- Faster convergence (fewer iterations).
- Parallelizable across scenarios.

Drawbacks

- Larger master problem (many cuts).
- Can be heavy with many scenarios.

Application Example

- Power systems: SCUC, stochastic OPF.
- Each contingency/scenario = one subproblem.
- Multi-cut Benders adds one cut per contingency.

Benders' decomposition – Stopping criterion

When can we consider that the algorithm has converged?

→ When the value of the **objective function of the “original problem”** in the subproblems and in the master problem has converged

We define at each iteration:

$$LB^{(i)} = f(x^{(i)}) + \alpha^{(i)}$$

$$UB^{(i)} = f(x^{(i)}) + \sum_{\omega} \pi_{\omega} g(y_{\omega}^{(i)}) \quad \leftarrow \text{The objective function of subproblem}$$

When $|LB^{(i)} - UB^{(i)}| \leq \varepsilon$, the algorithm has converged.

Stochastic market clearing: Benders' solution

- We assume that G_A can only deliver power to the day-ahead market
- The system operator commissioned wind power with the installed capacity of 80 MW

G_A	G_B	WP	Load
• Capacity: 100 MW • Production cost: 10 €/MWh • Inflexible generator	• Capacity: 40 MW • Production cost: 30 €/MWh • Flexible: reserve cap. Cost: 25 €/MW	• Capacity 80 MW • Production cost: 0 €/MWh	• 120 MW (inelastic and deterministic) • Rationing cost: 200 €/MWh

We assume that according to the best forecast for wind the realisation in a real-time production is 30 MW

Stochastic market clearing: Benders' solution

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

DA dispatch + allocated reserve capacity (first-stage decision variables)

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Mathematical formulation: Stochastic market-clearing

$$\underset{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \quad z = \underbrace{\sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d)}_{\text{Cost in the DA stage}} + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\begin{aligned} \sum_{k=1}^{N_G} p_k + w^{DA} &= l, \quad (\lambda^{DA}) \\ w^{DA} &\leq \bar{w} \\ p_k + R_k^u &\leq \bar{p}_k, \quad k = 1 \dots N_G \\ p_k - R_k^d &\geq 0, \quad k = 1 \dots N_G \end{aligned}$$

DA constraints

$$\begin{aligned} \sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) &= 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT}) \\ r_{k,s}^u &\leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s \\ r_{k,s}^d &\leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s \\ w_s^{SP} &\leq w_s^{RT}, \quad s = 1 \dots N_s \\ p_s^{rat} &\leq l, \quad s = 1 \dots N_s \end{aligned}$$

Mathematical formulation: Stochastic market-clearing

Real-time (RT) re-dispatch per scenario (second-stage decision variables)

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

Expected re-dispatch cost in the RT stage

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

RT constraints under scenarios s1 to s4

Stochastic market clearing: Benders' solution

Complicating variables: DA dispatch of generators and wind farm + allocated reserve capacity

$$\text{Minimize}_{p_k, R_k^u, R_k^d, w^{DA}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} z = \sum_{k=1}^{N_G} (c_k p_k + c_k^u R_k^u + c_k^d R_k^d) + \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} p_k + w^{DA} = l, \quad (\lambda^{DA})$$

$$w^{DA} \leq \bar{w}$$

$$p_k + R_k^u \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k - R_k^d \geq 0, \quad k = 1 \dots N_G$$

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^u, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^d, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

Stochastic market clearing: Benders' solution

Subproblem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}} \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right)$$

subject to

$$\sum_{k=1}^{N_G} (r_{k,s}^u - r_{k,s}^d) + p_s^{rat} + (w_s^{RT} - w^{DA(i)} - w_s^{sp}) = 0, \quad s = 1 \dots N_s \quad (\lambda_s^{RT})$$

$$r_{k,s}^u \leq R_k^{u(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$r_{k,s}^d \leq R_k^{d(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w_s^{sp} \leq w_s^{RT}, \quad s = 1 \dots N_s$$

$$p_s^{rat} \leq l, \quad s = 1 \dots N_s$$

$$R_k^{u(i)} = R_k^{u, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$R_k^{d(i)} = R_k^{d, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$w^{DA(i)} = w^{DA, \text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$p_k^{(i)} = p_k^{\text{fixed}(i)}, \quad k = 1 \dots N_G, \quad s = 1 \dots N_s$$

$$(\rho_{k,s}^{Ru(i)})$$

$$(\rho_{k,s}^{Rd(i)})$$

$$(\rho_{k,s}^{w(i)})$$

$$(\rho_{k,s}^{p(i)})$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to all complicating variables

Stochastic market clearing: Benders' solution

Master problem:

$$\underset{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}}{\text{Minimize}} \quad \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

Benders' cuts?

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\begin{aligned} \alpha^{(i)} \geq & \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right) \\ & + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} (R_k^{u(i)} - R_k^{u(j)}) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} (R_k^{d(i)} - R_k^{d(j)}) \\ & + \sum_{s=1}^{N_s} \rho_s^{w(j)} (w^{DA(i)} - w^{DA(j)}) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} (p_k^{(i)} - p_k^{(j)}), \quad j = 1, \dots, i-1 \end{aligned}$$

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to complicating variables R_k^u

$$- \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) - \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

$$+ \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) - \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Sensitivities of subproblem under scenarios s1 to s4,
with respect to complicating variables R_k^d

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\alpha^{(i)} \geq \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right)$$

$$+ \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right)$$

$$+ \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1$$

Sensitivities of subproblem under scenarios s1 to s4,
with respect to complicating variable w^{DA}

Stochastic market clearing: Benders' solution

Master problem:

$$\text{Minimize}_{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, \alpha^{(i)}} \sum_{k=1}^{N_G} \left(c_k p_k^{(i)} + c_k^u R_k^{u(i)} + c_k^d R_k^{d(i)} \right) + \alpha^{(i)}$$

subject to

$$\sum_{k=1}^{N_G} p_k^{(i)} + w^{DA(i)} = l, \quad (\lambda^{DA})$$

$$w^{DA(i)} \leq \bar{w}$$

$$p_k^{(i)} + R_k^{u(i)} \leq \bar{p}_k, \quad k = 1 \dots N_G$$

$$p_k^{(i)} - R_k^{d(i)} \geq 0, \quad k = 1 \dots N_G$$

$$\alpha^{(i)} \geq \alpha^{down}$$

$$\begin{aligned} \alpha^{(i)} \geq & \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k \left(r_{k,s}^u - r_{k,s}^d \right) + c^{rat} p_s^{rat} \right) \\ & + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Ru(j)} \left(R_k^{u(i)} - R_k^{u(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{Rd(j)} \left(R_k^{d(i)} - R_k^{d(j)} \right) \\ & + \sum_{s=1}^{N_s} \rho_s^{w(j)} \left(w^{DA(i)} - w^{DA(j)} \right) + \sum_{k=1}^{N_G} \sum_{s=1}^{N_s} \rho_{k,s}^{p(j)} \left(p_k^{(i)} - p_k^{(j)} \right), \quad j = 1, \dots, i-1 \end{aligned}$$

Sensitivities of subproblem under scenarios s1 to s4, with respect to complicating variable p_k

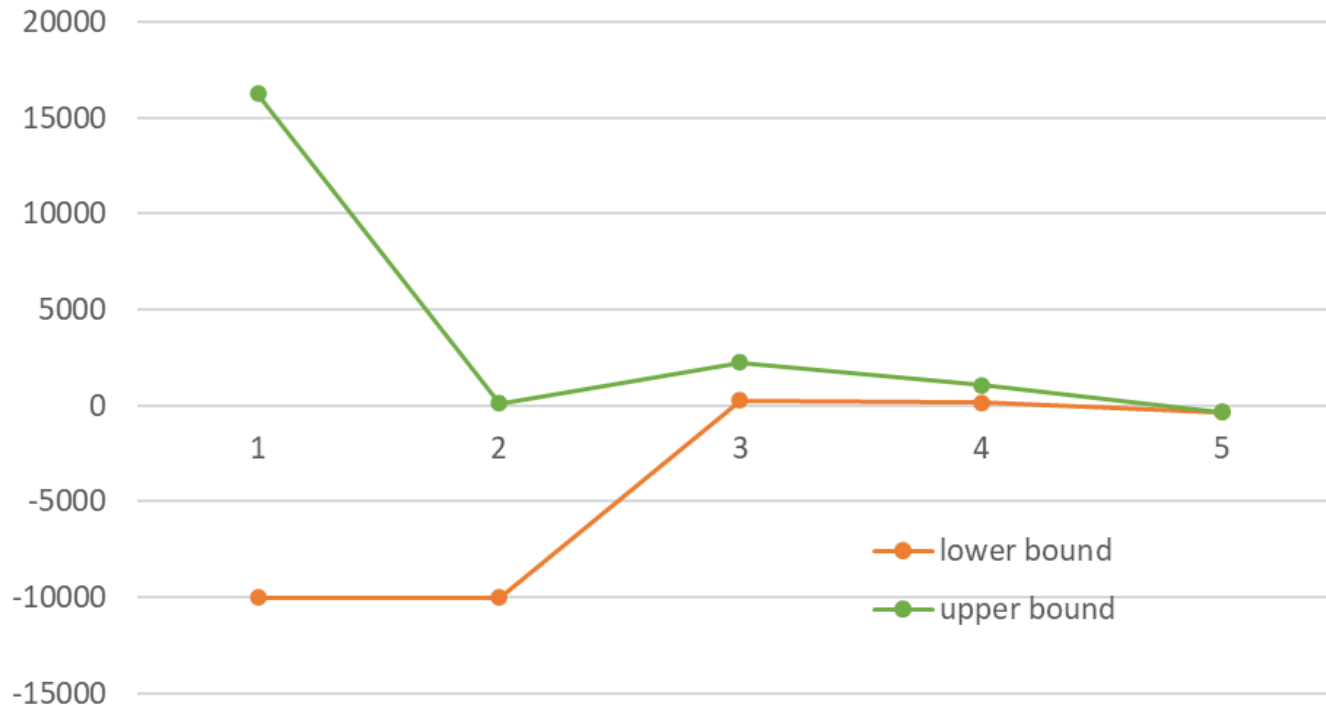
Stochastic market-clearing

	DA+R	RT (scenario 1) Wind power realization: 30 MW	RT (scenario 2) Wind power realization: 40 MW	RT (scenario 3) Wind power realization: 80 MW	RT (scenario 4) Wind power realization: 5 MW
G_A [MW]	100	unchanged	unchanged	unchanged	unchanged
G_B [MW]	15	-15 (down regulation)	-15 (down regulation)	-15 (down regulation)	unchanged
WP [MW]	5	+15 (10 MW spilled)	+15 (20 MW spilled)	+15 (60 MW spilled)	unchanged
R_B^u [MW]	0	-	-	-	-
R_B^d [MW]	15	-15	-15	-15	-
Load [MW]	120	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)	unchanged (not curtailed)
Corresponding operational cost [€]	1600	-450	-450	-450	0

Total expected operational cost of the system

$$= 1600 + 0.25 * [-450 - 450 - 450 + 0] = 1262.5 \text{ €}$$

Benders algorithm Convergence



$$\begin{aligned}
 UB^{(i)} &= \underset{p_k^{(i)}, R_k^{u(i)}, R_k^{d(i)}, w^{DA(i)}, r_{k,s}^u, r_{k,s}^d, w_s^{sp}, p_s^{rat}}{\text{Minimize}} \sum_{s=1}^{N_s} \pi_s \left(\sum_{k=1}^{N_G} c_k (r_{k,s}^u - r_{k,s}^d) + c^{rat} p_s^{rat} \right) \\
 LB^{(i)} &= \alpha^{(i)}
 \end{aligned}$$

...To sum up

- Stochastic optimization problems are decomposable
- Complicating variables: first-stage (here-and-now) decision variables
- Number of subproblems: one per scenario (at least)
- We have expressed the Benders' algorithm for a generic (linear) stochastic optimization problem
- We applied Benders' decomposition to stochastic market - clearing